FACE ATLAS: A transdiagnostic latent atlas of severe mental illness separates biological burden from clinical severity and forecasts functional outcome

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Abstract

Psychiatric diagnoses aggregate biologically heterogeneous patients, which limits mechanistic stratification and prognosis. We analysed the harmonized baseline assessment of $N = 9{,}013$ patients with bipolar disorder, schizophrenia or major depression from the three FACE cohorts using one global, missingness-aware Bayesian bifactor / exploratory structural-equation model, specified and fit to each patient's observed cells only (no imputation), with diagnosis withheld from model fitting. The model resolved a transdiagnostic coordinate system: a general functional-burden factor (G) and eight specific dimensions—cognition, metabolic, inflammatory, sleep, developmental risk, suicidality, mania and substance use. The central finding is a **dissociation between biology, symptoms and functional burden**: metabolic and inflammatory load were the *least* burden-entangled domains, and they remained so after adjustment for medication, adiposity and site. The coordinate space was a graded **continuum, not discrete biotypes**—apparent clustering was reproduced by a single-Gaussian null — yet four archetypal extremes gave a clinically legible reading, including a *biological* corner and a *severe, non-biological* corner with comparable functional burden but opposite biological profiles. Scored forward onto follow-up without re-estimation, biological coordinates were stable (test–retest intraclass correlation 0.91 for metabolic load) whereas symptom axes were state-like (they shifted with clinical state). The archetype phenotype predicted two-year functioning beyond diagnosis, baseline severity and baseline functioning (held-out expected log predictive density +59; two-year functional-remission 27%–60% across corners), but the individual-level discrimination gain was small (remission AUC +0.011). These results define a transdiagnostic biological atlas of severe mental illness that supports mechanistic stratification and group-level functional forecasting, and they delineate the requirements for clinical utility: external validation, incident-event outcomes and randomized treatment data.

Keywords: transdiagnostic psychiatry; dimensional psychopathology; bifactor model; metabolic syndrome; inflammation; bipolar disorder; schizophrenia; major depression; archetypal analysis; precision psychiatry.

1 Introduction

Psychiatric diagnoses are clinically indispensable but biologically nonspecific. The categories of the DSM aggregate patients who differ widely in their biology, course and treatment response, and they are undercut by pervasive comorbidity, within-category heterogeneity and diagnostic

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Table 1: **Sample characteristics by cohort (FACE baseline, N = 9,013).** Aggregate-only summary; sex coded female. Retention rows are the percentage of the baseline cohort re-assessed at the 12- and 24-month visits.

Characteristic	Bipolar	Schizophrenia	Depression	Overall
N	6,252	2,209	552	9,013
Age, years — mean (SD)	39.3 (13.4)	31.1 (9.6)	51.0 (13.3)	38.0 (13.4)
Age, years — median [IQR]	38 [28–49]	29 [23–37]	52 [42–60]	36 [27–48]
Female — n (%)	3,969 (63.5)	600 (27.2)	323 (58.5)	4,892 (54.3)
Education, years — mean (SD)	14.2 (2.8)	12.3 (2.6)	13.2 (3.2)	14.0 (2.8)
Recruitment sites — n	15	12	13	21
Education missing — n (%)	1,803 (28.8)	1,646 (74.5)	228 (41.3)	3,677 (40.8)
DSM-5 subtype — n (% of cohort)				
Bipolar I	2,635 (42.1)	—	—	
Bipolar II	2,956 (47.3)	—	—	
Bipolar, unspecified	661 (10.6)	—	—	
Schizophrenia	—	1,692 (76.6)	—	
Schizoaffective disorder	—	476 (21.5)	—	
Schizophreniform disorder	—	41 (1.9)	—	
Major depressive disorder	—	—	552 (100.0)	
Retained at 12 months — %	49.2	43.6	42.2	47.4
Retained at 24 months — %	35.6	26.9	24.5	32.8

instability over time [1, 2]. Two influential programmes have argued in response that psychopathology is better described by continuous, *transdiagnostic* dimensions than by categorical boundaries: the U.S. National Institute of Mental Health's Research Domain Criteria (RDoC) [1] and the Hierarchical Taxonomy of Psychopathology (HiTOP) [2, 3]. A recurring empirical motif within this work is a single general factor that spans disorders—the "p factor"—most defensibly read as a dimension of life impairment rather than a latent liability to specific symptoms [4, 5], although the substantive meaning of such general factors remains contested [6–8].

This dimensional structure has, however, been mapped almost entirely from a single *kind* of measurement. Symptom-based work [2, 4] and brain-imaging work [9–11] each describe one slice of the patient. The routinely collected *biology* that dominates the physical-health burden of severe mental illness—the metabolic and inflammatory markers that drive its excess mortality—has not been placed *inside* the transdiagnostic factor space across disorders, even though an independent literature suggests it forms a partly separable axis [12, 13]. Metabolic and inflammatory abnormality co-occur in a sizeable minority of mood-disorder patients and track atypical, energy-related symptoms rather than overall severity [12, 14–16], and metabolic burden in severe mental illness follows medication, adiposity and chronicity more than symptom load [17–20]. Whether biological load is genuinely decoupled from a transdiagnostic severity axis, and by how much, has never to our knowledge been measured directly.

A second question is structural, and it determines what a clinician would do with such a map. If patients vary continuously, the actionable object is a position on an axis; if they fall into discrete biological kinds, it is group membership. Normative-modelling studies of brain structure in these disorders report heterogeneity so extensive that "the average patient" is barely informative [9–11]. Against this, the B-SNIP programme has identified replicable, brain-based psychosis "biotypes" that cut across DSM boundaries [21–23]. The continuum-versus-biotype question is therefore genuinely open, and its answer may depend on which measurement space is interrogated [24].

Here we address both questions in a single analysis, and then ask how far the resulting map can be pushed. Why represent patients by latent dimensions rather than the raw measurements? Routine clinical and biological data are high-dimensional, redundant and pervasively

Table 2: **The nine-dimension transdiagnostic map.** Anchoring indicators, entanglement with general burden, adjudication verdict and cross-cohort measurement invariance for each dimension. Correlation with G is from the correlated-G model (continuous backbone); for mania and substance it is the bifactor direct $|\lambda_G|$. Depression/anxiety instruments are cross-loading windows on G (loadings 0.66–0.80), not a separate factor; anhedonia was rejected; impulsivity, negative symptoms and sensory abnormality were not testable for lack of common indicators.

Dimension	Anchoring indicators	corr G	Verdict	Invariance
G (burden)	FAST, EGF, EQ-5D (no symptoms)	—	severity factor	partial (SZ)
Cognition	TMT, WAIS coding, fluency	0.39	confirmed	invariant
Metabolic	BMI, lipids, glycaemia, BP	0.12	confirmed (split)	invariant
Inflammatory	CRP, leukocyte subsets	0.07	confirmed (split)	re-weights (DR)
Sleep	PSQI total + subscales	0.42	confirmed	invariant
Developmental risk	CTQ, perinatal, family history	—	confirmed (proxy)	invariant
Suicidality	ISF ideation / attempt (binary)	—	confirmed (mixed)	3-cohort
Mania	YMRS, Altman	0.15	confirmed	Altman floors (DR)
Substance	alcohol/cannabis SUD, nicotine	0.13	confirmed	BP–SZ (2-cohort)

incomplete; a latent model pools many noisy, overlapping indicators into a few reliable axes, places heterogeneous data types (laboratory values, ordinal scales, counts) on one comparable scale, and—fit to observed cells only—gives every patient a position with quantified uncertainty rather than discarding incomplete records, a compression we later show to be a near-sufficient representation of the raw indicators for prognosis. Using the harmonized baseline of the three FACE cohorts—bipolar disorder, schizophrenia and major depression, $N = 9{,}013$—we build *FACE-ATLAS*: one global, missingness-aware Bayesian sparse bifactor / exploratory structural-equation model with mixed likelihoods, fit to each patient's observed cells only, that places clinical, cognitive, behavioural and laboratory measurements in one coordinate system while holding diagnosis out as metadata. We then interrogate the fitted map through five dependent, increasingly demanding questions: does it *exist* as a stable structure; does it *organize* patients into biotypes or a continuum; does it *persist* over two years of follow-up; does it *predict* outcomes beyond diagnosis and severity; and—this being an observational, treatment-as-usual cohort—how far it can *inform treatment* before randomized data would be required? The answers are, respectively: a nine-dimension map on which metabolic and inflammatory load are the domains least entangled with functional severity; a graded continuum rather than discrete biotypes, legible through four archetypal extremes; a geometry that persists, with durable biology and moving symptoms; a real but modest, group-level prognostic signal for functioning carried by the biological phenotype; and a treatment analysis that is deliberately bounded—marking where observational, expert-centre data reach their limit and specifying the randomized evidence that establishing treatment selection would require. We report the modest and the null results as deliberately as the positive ones. The contribution is a calibrated atlas that separates biological burden from clinical severity and forecasts functional course at the group level, not an overstated one; that calibration is itself the antidote to a literature of biotype and biomarker claims that have repeatedly failed to replicate.

Methodologically, FACE-ATLAS specifies a domain-tailored Bayesian bifactor / exploratory structural-equation model for severe mental illness, rather than simply applying a generic factor-analysis template. It builds on established latent-variable tools [25–27] but advances their use by binding them into a single diagnosis-held-out, uncertainty-propagating atlas: mixed and

Gaussian-copula observation layers, an observed-cell likelihood for structured missingness, a correlated-G sensitivity arm for the biology–severity estimand, an archetypal representation of a continuous patient space, and downstream errors-in-variables prognosis and treatment-moderation analyses. The novelty lies in the specification, estimand and validation chain: routine biological load is placed *inside* a transdiagnostic psychiatric factor space, its separation from functional burden is quantified rather than assumed, and its temporal and prognostic implications are tested without converting a continuum into false biotypes. At the model level, this specification combines a psychiatric construct-ontology prior with per-indicator likelihoods for mixed clinical and biological data, and we derive the model's observed-data likelihood and full-information evaluation under structured missingness. The derivation is for this configured FACE-ATLAS model, not for a new generic estimator or likelihood family. It makes the inferential contract auditable: standard factor-analysis identities, observed-data marginalization and the Woodbury evaluation are assembled because together they define the atlas—diagnosis is not an input, missing cells are not imputed, G–specific correlations are measured in a separate correlated-G arm, and patient-score uncertainty is propagated into stratification, prognosis and treatment analysis.

2 Results

2.1 Sample and observed-data modelling population

The analysis sample was the harmonized baseline (visit V0) of the three FACE cohorts: $N = 9{,}013$ patients (bipolar disorder [BP] 6,252; schizophrenia [SZ] 2,209; major depression [DR] 552) recruited across 21 expert-centre sites (Table 1). The cohorts differ in age, sex distribution and chronicity, as expected for these disorders. We deliberately fit the measurement model on the *full* sample without completeness selection. Of the 201 usable harmonized common variables, 143 entered the model as indicators (mean cell missingness 39.8%, preserved as missing and never imputed); the remaining 58 are identifiers, covariates and validation labels (ambiguous-direction laboratory values, treatment-efficacy ratings, ECG intervals, reproductive/hormonal and smoking-history confounders) and were never modelled as indicators. Diagnosis (cohort and DSM-5 subtype) was held out of the model and used only as a validation or metadata label. The reported FACE-ATLAS map is one global, cohort-reweighted, full-sample fit estimated with a Gaussian-copula (rank-based) marginal model; because the same structure is recovered under a plain Gaussian likelihood, the central findings do not depend on that modelling choice—a robustness result, not an assumption (Annex A). Follow-up visits at 12 and 24 months (retained: 47.4% and 32.8% of V0; Extended Data Fig. F3) were used only to test temporal coherence and prognosis, never to define the map. The high-level model is summarized below and in Methods; the full derivation of its observed-data likelihood—the rotation and identification argument, the covariance identity and the equivalence with full-information maximum likelihood—is given in Annex A. Figure 1 presents the atlas at a glance; the study design and the three load-bearing invariants are detailed in Extended Data Fig. F1.

2.2 The fitted object: a posterior coordinate for each patient

The primary output of FACE-ATLAS is not a cluster label but a *posterior coordinate distribution*. For patient i the model returns a nine-dimensional coordinate

$$f_i = (G_i,\ D_{i,\mathrm{cog}},\ D_{i,\mathrm{met}},\ D_{i,\mathrm{inf}},\ D_{i,\mathrm{sleep}},\ D_{i,\mathrm{dev}},\ D_{i,\mathrm{suic}},\ D_{i,\mathrm{mania}},\ D_{i,\mathrm{subst}}) \in \mathbb{R}^9, \qquad (1)$$

one general functional-burden coordinate G_i and eight specific coordinates (cognition, metabolic, inflammatory, sleep, developmental risk, suicidality, mania, substance use)—the eight specific axes are established as distinct, data-earned dimensions in §2.3 (Table 2). What the model

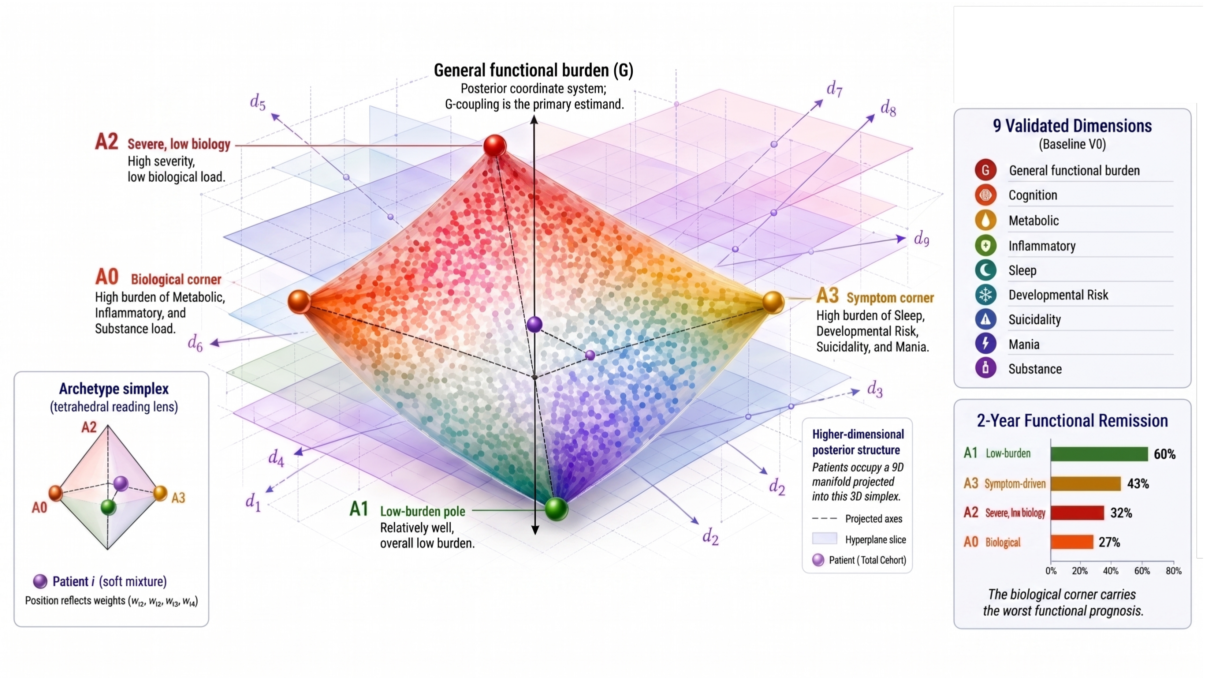

Figure 1: **Illustration of the FACE ATLAS: a latent dimensions of severe mental illness and corresponding archetypes.** The global, missingness-aware Bayesian bifactor / exploratory structural-equation model, fit to the FACE baseline ($N = 9{,}013$) with diagnosis held out, yields a nine-dimension transdiagnostic latent space—a general functional-burden factor (G) and eight specific dimensions. The patient cloud is a graded continuum (best silhouette 0.140, indistinguishable from a single-Gaussian null 0.141 ± 0.002), read through four archetypal extremes whose convex blends describe every patient: a *biological* corner (A0; high metabolic, inflammatory and substance load), a *low-burden* pole (A1), a *severe, non-biological* corner (A2), and a *symptom-driven* corner (A3). Metabolic and inflammatory load are largely dissociated from general burden; biology is durable; the archetype phenotype improves held-out two-year functional prediction (ΔELPD $+59.4 \pm 11.1$); and two-year functional remission ranges 27%–60% across corners, the biological corner worst. A0 and A2 carry comparable severity but opposite biology.

actually delivers is the posterior of this coordinate given the patient's *observed* indicators x_{i,O_i} alone—no imputation, no completed data matrix (Methods):

$$p(f_i \mid x_{i,O_i}) \;\approx\; \mathcal{N}\Big(\hat{f}_i,\; S_i\Big)\,, \tag{2}$$

whose mean $\hat{f}_i$ is the patient's *position* and whose covariance S_i is the *uncertainty* of that position. A patient is therefore represented by a location with error bars, not by a category. Two patients can share the same estimated metabolic score yet differ in its reliability—one with a complete laboratory panel, the other with a single observed marker—and the model records that difference: each coordinate carries a posterior mean, a standard deviation, a highest-density interval, the number of contributing indicators and a reliability tier. Every downstream analysis in this paper (structure tests, archetype weights, temporal decomposition, prognosis and treatment) consumes the full $(\hat{f}_i, S_i)$ rather than treating coordinates as exact, so a sparsely measured patient is never mistaken for a precisely located one. *Clinically,* this fixes the unit of analysis: the question is not "which biotype is this patient?" but "where does this patient sit on nine continuous axes, and how precisely do we know it?"

2.3 The nine dimensions of the transdiagnostic atlas

From ten candidate constructs, one global model returned a stable nine-dimension structure: a general functional-burden factor (G) plus eight weakly correlated specific axes—cognition, metabolic, inflammatory, sleep, developmental risk, suicidality, mania and substance use (Figure 2; Table 2). Inter-factor correlations among the specific axes were weak throughout (mean absolute off-diagonal ≈ 0.08; Figure 3b), confirming genuinely distinct axes rather than one collapsed dimension. G was anchored only by functioning and global-severity indicators (FAST loading 0.91, EGF 0.64) and carried no symptom content, supporting its conservative reading as transdiagnostic *functional burden* rather than a symptom-liability "p" factor. Depression and anxiety instruments (MADRS, QIDS, STAI) did not form a separate factor; they loaded 0.61–0.82 on G as cross-loading "windows" onto burden. Of the ten prior candidate factors, nine were confirmed and one—anhedonia—was rejected (absorbed into G and the depression/anxiety windows); the metabolic and inflammatory axes, entered a priori as a hypothesised split of cardiometabolic and immune load, were confirmed as two distinct factors (inter-factor correlation ≈ 0.18) rather than one. A further three constructs with no common indicator—impulsivity, negative symptoms and sensory abnormality—were declared *not testable*, each verdict an outcome of the analysis rather than an assumption (Table 2). The structure was not an artefact of the priors: a flat-prior refit reproduced the loadings and factor correlations essentially exactly (Tucker congruence $\varphi = 1.00$), and model comparison decisively preferred the bifactor over unidimensional and correlated-factor alternatives. Absolute fit was acceptable across both likelihood blocks (continuous Bayesian standardized root-mean-square residual ≈ 0.07; 21/22 non-Gaussian indicators reproduced within the 90% posterior-predictive interval). *Clinically,* the axes answer two different questions: G answers *how impaired* the patient is, while the eight specific axes answer *what kind* of burden the patient carries.

2.4 Biology is separable from symptoms and severity (the central result)

Because the bifactor identification fixes $G \perp$ specifics by construction, we re-estimated the model allowing G to correlate freely with the specific axes. The resulting vector of general–specific correlations, $\phi_{Gd} = \mathrm{Corr}(G, D)$ (Methods, Eq. 8), is the *primary estimand* of FACE-ATLAS: it converts "biology is separable from severity" from a modelling constraint into a measured quantity. Under this correlated-G model, metabolic and inflammatory load were the *least* entangled with functional burden: correlations with G of 0.12 (metabolic) and 0.07 (inflammatory), against 0.39 for cognition and 0.42 for sleep (Figure 4). In shared-variance terms the gap is stark—metabolic and inflammatory load share only $\approx 1.4\%$ and $\approx 0.5\%$ of their variance with general burden, against $\approx 15\%$ (cognition) and $\approx 18\%$ (sleep). And because the specific axes are themselves only weakly inter-correlated (mean absolute off-diagonal ≈ 0.08), biological load is separable not just from severity but from the symptom axes too: biology $\perp$ symptoms $\perp$ severity. Two patients with the same overall impairment can therefore carry very different metabolic and immune loads—a patient's biological burden rides largely on axes that the global clinical-severity picture does not capture. This dissociation was robust to the obvious confounders. Adjusting each indicator for age, sex, education and recruitment site, and then additionally for antipsychotic exposure, did *not* raise the biological correlations—if anything it lowered them (metabolic $0.12 \rightarrow 0.07$, inflammatory $0.07 \rightarrow 0.06$), and the biological axes remained the least entangled at every step (Figure 4; Annex A). Consistent with a large independent literature (Discussion), we state this as relative, not absolute, independence: metabolic and inflammatory load are *largely independent of a general functional-burden axis*, with the 0.07–0.12 values quantifying—at the low end of what that literature would predict—a pattern previously described only qualitatively. *Clinically,* this means two patients with comparable functional burden can have sharply different cardiometabolic or inflammatory profiles: severity cannot be used as a proxy for biological load. This is the property a biology-aware stratification can exploit, and it is the load-bearing premise

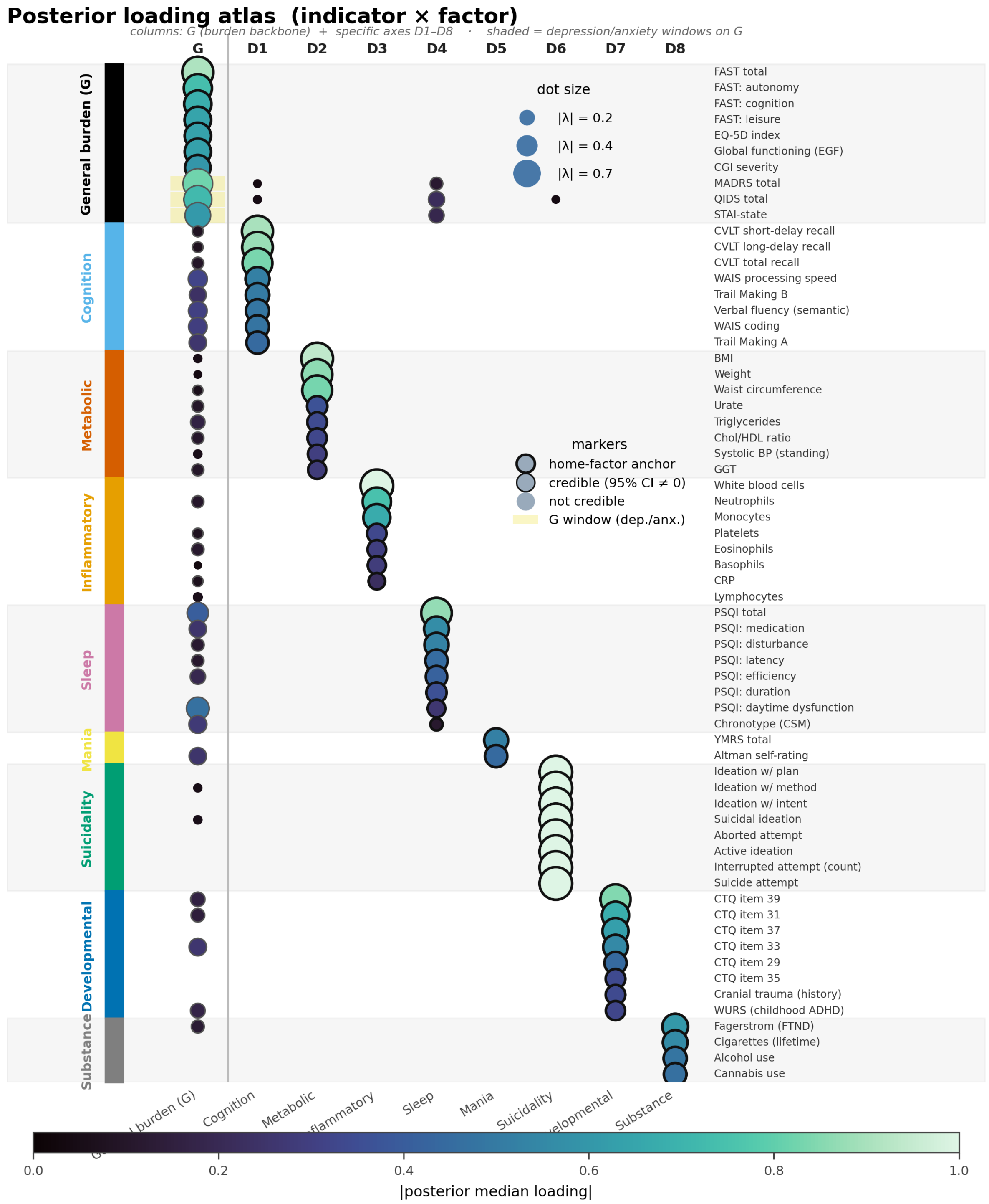

Figure 2: **Posterior loading atlas of the measurement model.** Rows are clinical indicators grouped by their home factor (coloured left strip and alternating bands); columns are the general functional-burden dimension G and the eight specific latent axes (D1–D8). Dot size and colour encode the absolute posterior median loading; outlined dots indicate loadings whose 95% credible interval excludes zero, and a heavier ring marks the home-factor anchor. Specific axes are each anchored by a coherent clinical indicator block, whereas G captures broad functional burden and carries cross-loading "windows" over the depression and anxiety instruments (MADRS, QIDS, STAI; shaded on the G column). The full nine-factor map is shown (continuous backbone plus the mania and substance axes from the joint mixed fit); the complete atlas over all modelled indicators is Extended Data Fig. F2. This structure supports an atlas-like reading of the latent space: specific dimensions are clinically localised axes, while G acts as a transdiagnostic burden backbone. Factor-wise interpretability and the inter-factor correlation matrix are in Figure 3.

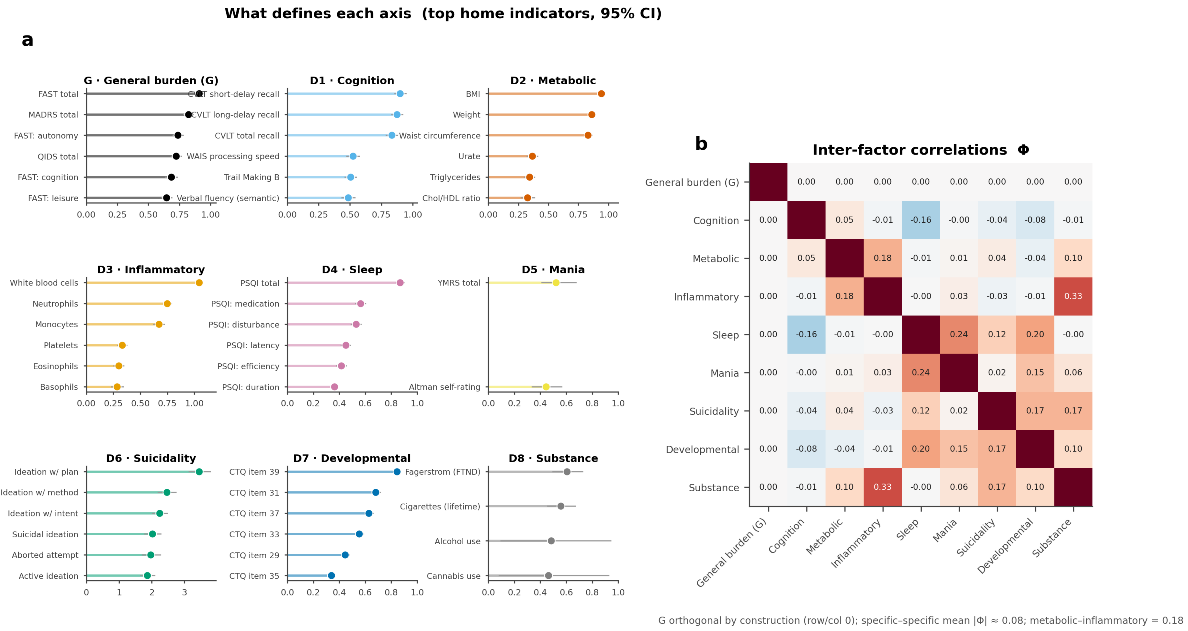

Figure 3: **Factor interpretability and inter-factor correlations.** (**a**) Factor-wise lollipops: for each axis (G and D1–D8), its strongest home indicators with 95% credible whiskers—"what defines each axis." (**b**) The inter-factor correlation matrix Φ is weak throughout (specific–specific mean |off-diagonal| ≈ 0.08); G is orthogonal to the specific axes by construction (top row and column), and metabolic and inflammatory load correlate only ≈ 0.18. The one larger specific cell, inflammatory–substance ≈ 0.33, reflects substance's less-identifiable correlations (modelled as an independent axis downstream); the biology×symptom cells all stay at the off-diagonal floor.

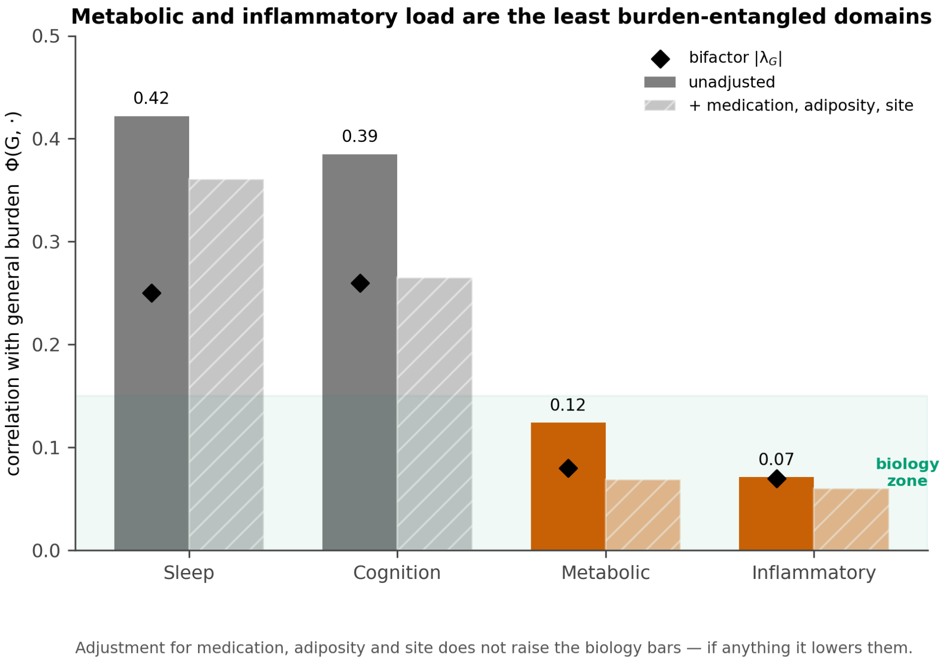

Figure 4: **Metabolic and inflammatory load are the least burden-entangled domains.** Correlation of each continuous specific axis with the general functional-burden factor G (bars), unadjusted and after adjustment for medication, adiposity and site; the bifactor general loading $|\lambda_G|$ (diamonds) is shown for comparison. Metabolic and inflammatory load fall in the "biology zone" ($\Phi < 0.15$) and stay there under adjustment, whereas cognition and sleep partly track G.

of everything that follows.

2.5 The coordinate cloud is continuous, not discretely clustered

We next asked whether these nine coordinates fall into separable kinds. They do not—a *negative structural result*. On the cloud $\{(\hat{f}_i, S_i)\}$, a structure-discovery battery run over the posterior draws—cluster tendency, modality, silhouette, the gap statistic, a Gaussian-mixture information criterion, density-based clustering and a topological summary—returned a continuum verdict: the silhouette peaked at only 0.14 (below the 0.15 separation floor), the first principal component was unimodal (Hartigan dip $p \approx 1.0$), density-based clustering assigned 100% of points to noise, and the topological summary was a single connected component. A single-Gaussian *falsification null* made the verdict decisive: the best partition of the real cloud separated patients no better than slicing a structureless Gaussian blob of identical covariance (best silhouette 0.140 for the real data versus 0.141 ± 0.002 for the null, $z = -0.36$; the Gaussian-mixture-optimal number of components collapsed to one in all 20 posterior draws; Figure 5a). There are, in this clinical–biological space, no well-separated, reproducible discrete clusters.

The continuum is genuine and transdiagnostic, not an artefact. Diagnosis is intermixed throughout the cloud (adjusted Rand index ≈ 0 against both cohort and the seven DSM-5 subtypes; Figure 5b); overall burden forms a smooth gradient along the principal axis (Figure 5c); and inflammatory load varies along a *different* direction (Figure 5d)—the biology $\perp$ G dissociation, made visible. The verdict is not a missingness artefact (a classifier predicting position from each patient's coverage pattern performed below baseline, lift -0.05, permutation $p = 0.23$) and is a tighter description of the coordinate cloud than diagnosis (free-mixture information criterion 197,900 versus 201,300 for a DSM-5-constrained model). These same properties explain why clustering the *raw* indicators can appear to yield biotypes where the latent space does

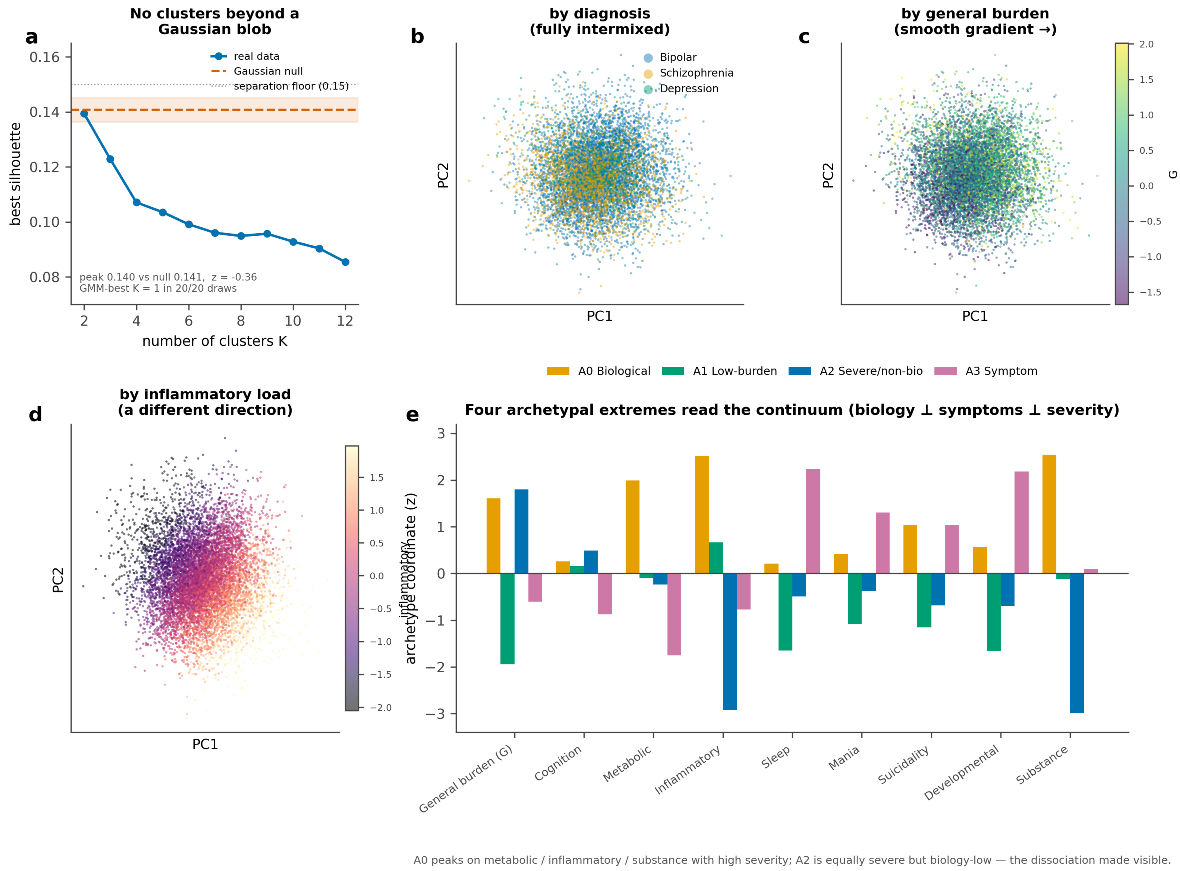

Figure 5: **The patient space is a continuum, not biotypes.** (**a**) A single-Gaussian falsification null: the best silhouette achievable on the real nine-dimensional cloud is no better than on a structureless Gaussian of matched covariance ($z = -0.36$); the mixture-optimal number of clusters is one in all posterior draws. (**b–d**) Principal-component embedding of the coordinates ($N = 9{,}013$), coloured by diagnosis (fully intermixed), by general burden G (a smooth gradient), and by inflammatory load (a gradient in a *different* direction).

not [24]: a partition of routine clinical data is driven largely by measurement-side structure rather than patient biology—the structured missingness and measurement coverage just shown to be uninformative here (which panels each cohort and site actually ran), a handful of high-variance, heavy-tailed laboratory variables, blocks of redundant correlated indicators, and cohort- and site-specific instruments and units that re-encode diagnosis. The measurement model neutralizes each of these—an observed-cell likelihood instead of imputation, per-indicator and Gaussian-copula likelihoods instead of raw scales, a few shared factors instead of redundant indicators, and diagnosis held out under measurement invariance—so the structure that remains is genuinely transdiagnostic and, unlike a raw partition, is tested against a structureless null rather than assumed. *Clinically,* the appropriate object is therefore a coordinate with uncertainty (Eq. 2), not a categorical biotype.

2.6 Four archetypal extremes define the shape of the continuum

A continuum has no clusters, but it still has a *shape.* Archetypal analysis recovers that shape as a small set of reproducible extreme corners—a *positive geometric result* that coexists with the negative clustering result above. Each patient is written as a convex blend of $A = 4$ extreme

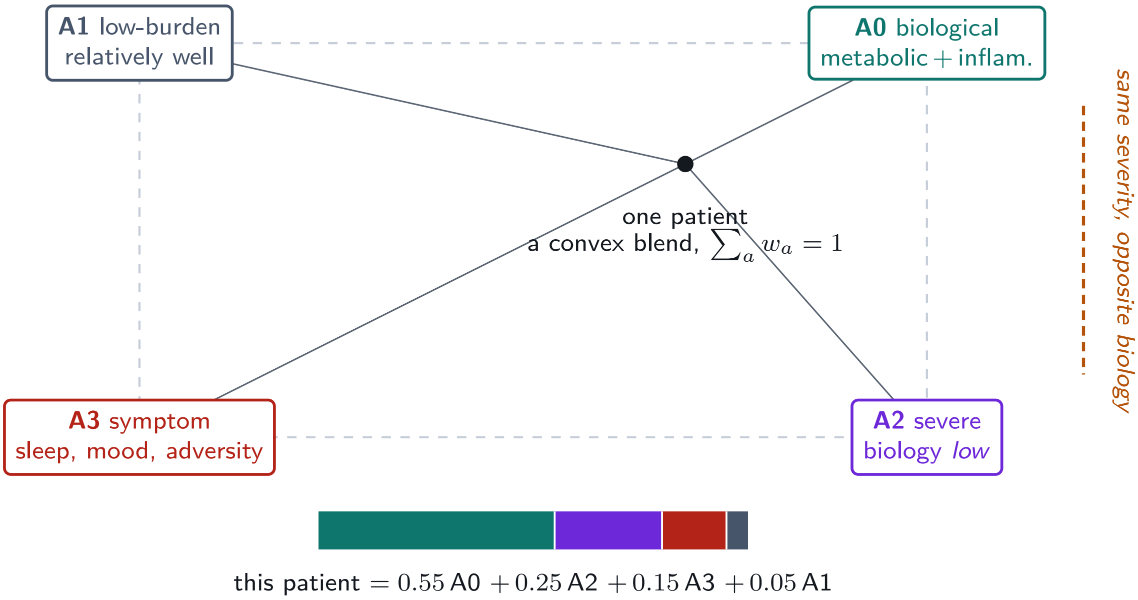

Figure 6: **The archetype simplex: every patient a convex blend of four extremes.** The four archetypal extremes are the corners—**A0** biological (cardiometabolic + inflammatory + substance), **A1** low-burden, **A2** severe-but-low-biology, and **A3** symptom-driven—and every patient sits *inside* the simplex as a convex blend whose barycentric weights w_i sum to one (worked example, foot). The load-bearing dissociation is geometric: **A0** and **A2** are distinct corners at essentially the *same* general burden but opposite biology—biology $\perp$ severity, rendered as a direction one can point to.

phenotypes,

$$\hat{f}_i \approx \sum_{a=0}^{3} w_{ia}\, z_a, \qquad w_{ia} \geq 0, \qquad \sum_{a=0}^{3} w_{ia} = 1. \tag{3}$$

The four z_a are extreme points on the convex hull of the patient cloud, and the weights w_{ia} are *barycentric coordinates*: they quantify how strongly each patient is pulled toward each extreme. Dominant-archetype labels are used only for summaries; the inferential object is the continuous weight vector w_i, not a class assignment. The number of corners ($A = 4$) was set by cross-seed stability rather than by a parsimony rule—only two- and four-corner solutions reproduced across seeds (cross-seed Tucker congruence 0.98)—and uncertainty was propagated by refitting Eq. (3) across posterior draws (Annex B).

The four extremes separate biology from symptoms from severity (Figure 6): a *biological* corner (A0; high metabolic, inflammatory and substance load), a *low-burden* pole (A1), a *severe, non-biological* corner (A2; high overall severity but low biological load), and a *symptom* corner (A3; high sleep, developmental, suicidality and mania). The biological corner A0 and the severe corner A2 carry comparable general burden but opposite biological profiles—A0 versus A2 is precisely why *severity is not biology*, the central dissociation now rendered as geometry. Patients sit *between* the corners far more than at them: the weight vectors are diffuse (mean entropy 0.78 across the four corners), and only 39% of patients have a single dominant corner. Critically, because biological load varies continuously without a density *gap*, it never forms a clean partition boundary at any granularity; it surfaces instead as a *direction of maximal phenotype*—an archetypal corner—which is exactly why the continuous coordinates and the weight vectors w_i, not any hard tessellation, are the load-bearing objects downstream. *Clinically,* the archetypes provide a vocabulary for reading the continuum, not a replacement for it.

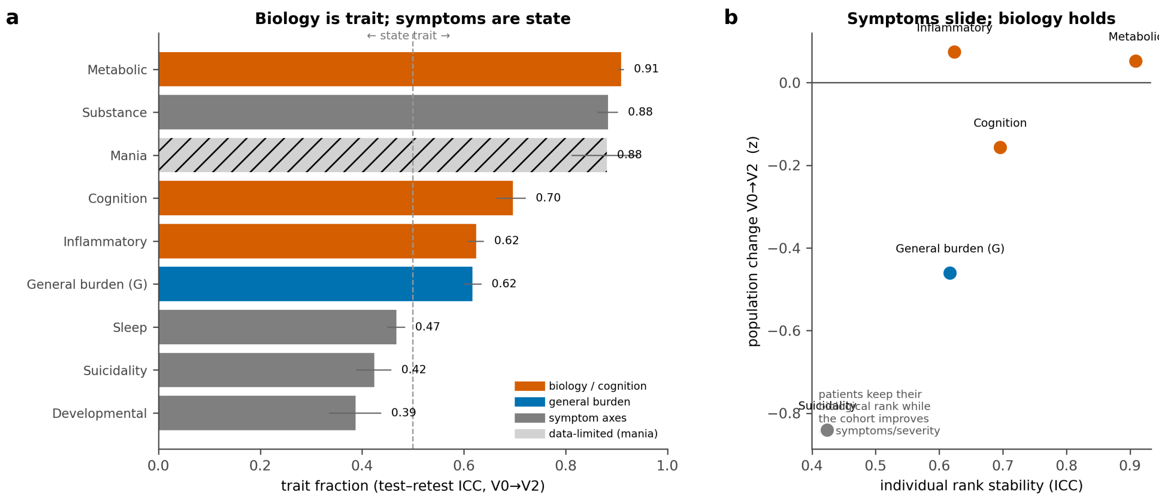

Figure 7: **Durable biology, moving symptoms.** (**a**) Trait fraction (test–retest intraclass correlation, with 94% credible intervals) over follow-up: the biological and cognitive axes are trait-like, the symptom axes state-like; mania is data-limited (hatched). (**b**) Individual rank stability versus population-level change: patients keep their biological rank (high intraclass correlation, near-zero population shift) while the cohort slides down the severity and symptom axes.

2.7 The geometry persists over two years (the map persists)

Scored forward onto the 12- and 24-month visits on the *fixed* model (observed cells, uncertainty propagated, never re-estimated; baseline coordinates reproduced at Pearson $r \approx 0.99$ through this path), the measurement remained invariant for all five testable backbone axes (longitudinal Tucker congruence $\varphi = 0.97$–1.00; inflammatory load, only partially invariant on a plain Gaussian map, was invariant here at $\varphi = 0.97$; Extended Data Fig. F4). Coordinate change can therefore be read as patient change rather than instrument drift. A measurement-error decomposition then separated trait from state (Figure 7). The durable, trait-like axes were the biological and cognitive ones (test–retest intraclass correlation: metabolic 0.91, substance 0.88, cognition 0.70, inflammatory 0.62); the moving, state-like axes were the symptom ones (sleep 0.47, suicidality 0.42, developmental risk 0.39). General burden was trait *by rank* (intraclass correlation 0.62): patients largely kept their relative position even as the cohort as a whole improved on severity and symptoms (population shift -0.46 on G and -0.84 on suicidality, while biology stayed static). The archetype geometry persisted—each patient's weight vector w_i moved little over two years (median cosine similarity 0.90). *Clinically,* the reading is compact: *stratify on the durable biology; monitor the moving symptoms.*

2.8 Archetype position predicts group-level two-year functioning (the map predicts)

We then asked whether a baseline coordinate or phenotype predicts a two-year outcome *incrementally beyond* diagnosis, current severity and the baseline value of the outcome, using an errors-in-variables Bayesian model that propagated each coordinate's measurement uncertainty. The map added predictive signal for *functioning* but not for severity (which was saturated by its own baseline autoregression). The signal was carried by the *phenotype*, not by any single laboratory axis: the four archetypal extremes added the most held-out predictive value (expected log predictive density gain +59 over the diagnosis-plus-severity-plus-baseline reference), the eight specific axes added +37, and—an honest, map-specific shift from our earlier analyses—the isolated "durable biology" axes *alone* no longer cleared the bar (+2, ambiguous; Figure 8b).

Biology still predicts functioning, but as a *corner of the phenotype* (the biological archetype) rather than as a standalone three-axis block. No hard partition improved on the continuous and archetype encodings (every tessellation added $\approx +20$, strictly dominated by the archetypes); the operative number of clusters is therefore *none*—the actionable object is a continuous position, not a category. The map was *co-informative* with DSM-5 rather than redundant with it (functioning: map +22, DSM-5 +29, both +67; Figure 8c), and *course-dependent*, with the signal concentrated in the episodic, open-course cohorts (bipolar disorder and depression) and weak when bipolar disorder was removed.

Translated to the clinic, ranking patients by their dominant archetypal extreme yielded two-year functional-remission rates from 27% to 60%, transdiagnostically (Figure 8a). The *biological* corner A0 carried the worst functional prognosis (27%) and the low-burden pole A1 the best (60%): A0 versus A1 is why the biological corner *matters prognostically.* A0 also remained worse than the equally severe non-biological corner A2 (27% vs 32%)—the severity-is-not-biology dissociation, now translated into outcome. This gradient was a genuine within-diagnosis effect, not a cohort-composition artefact: the rank held inside every cohort (a biological-vs-low-burden odds ratio ≈ 4 in each), standardizing to a common cohort mix moved it by only $\approx 6\%$, and the corner-by-cohort interaction was non-significant. The individual-level discrimination gain was nonetheless small (functional-remission AUC +0.011 over a strong diagnosis-plus-severity reference). These are not in tension: a wide spread in *group* remission rates with a small *individual* AUC gain is the expected signature of groups that differ in average risk but overlap substantially at the patient level, layered on an already-strong autoregressive baseline—the map enriches and forecasts at the group level without sharply separating individuals. The functioning signal survived inverse-probability weighting for attrition (+53), restriction to either two of the three cohorts (+56, +59), and a permutation null (which correctly vanished). Finally, against the full set of 143 raw indicators under a matched gradient-boosting classifier, the nine-dimension map was a *sufficient* representation for predicting deterioration (equal AUC) and *near-sufficient* for recovery (raw indicators +0.04 AUC), with 97% of the raw recovery signal already contained within the nine factors (Extended Data Fig. F5)—the compression is faithful, not lossy in any axis the map omits. *Clinically,* the atlas is useful for group-level enrichment and monitoring, not as an individual remission calculator.

2.9 Treatment moderation was not detected in observational treatment-as-usual

Treatment exposures, recovered from the per-cohort source dictionaries and harmonized to drug-class indicators, were run through an identification-first causal pipeline (common-support overlap gate → propensity weighting → doubly-robust errors-in-variables moderation with an E-value sensitivity analysis and a minimum-detectable-effect calculation) to ask whether the durable axes or the archetypal extremes *moderate* treatment response. On observational, expert-centre treatment-as-usual they did not reliably do so (Extended Data Fig. F6a). Where identification was cleanest—lithium in bipolar disorder, near-complete overlap—the result was a *well-identified, bounded null*: the average effect was indistinguishable from zero (E-value 1.06) and the design could have resolved an interaction as small as ≈ 0.2 standard deviations but did not. An antipsychotic-by-biology signal for functioning in bipolar disorder was a confounded *average* effect (E-value 1.77) with *no* reliable moderation (the two map encodings disagreed on which axis carried it—the signature of a false positive); and clozapine in schizophrenia was channelled to treatment-resistant patients and hence underpowered rather than cleanly null. Average treatment effects were uniformly confounding-fragile (E-values 1.1–1.8), the expected signature of confounding by indication.

This boundary is earned, not assumed, and it cuts two ways that strengthen the rest of the paper. First, the prognostic finding was *not* merely unmodelled treatment: the archetype carrier of the functioning forecast survived adjustment for the drug classes patients were actually

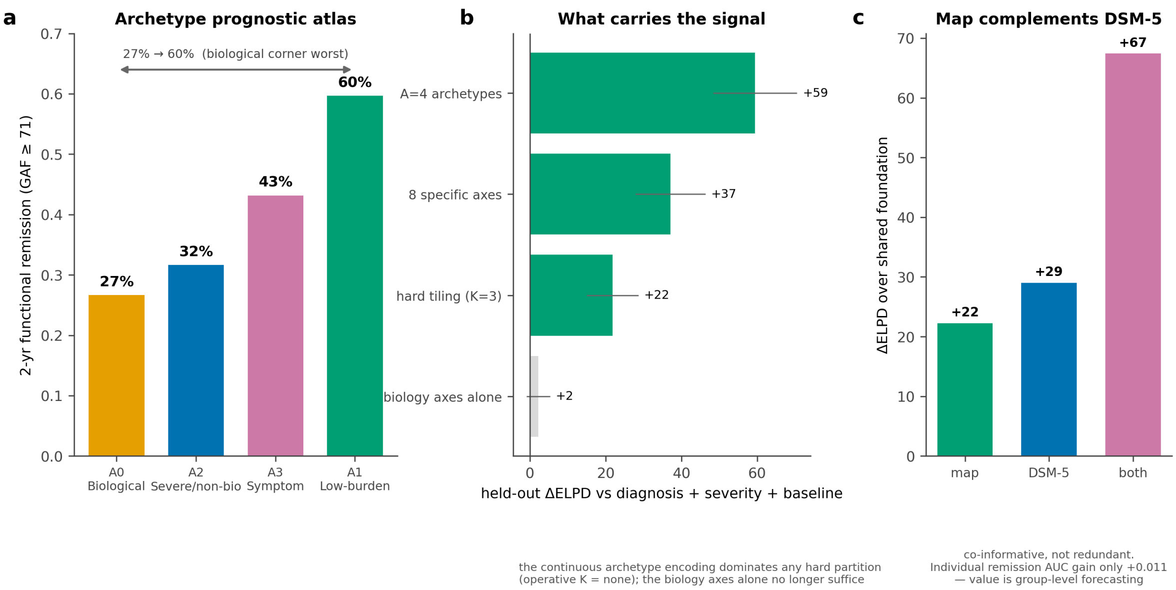

Figure 8: **Prognostic reach for functioning.** (**a**) Archetype prognostic atlas: two-year functional-remission rate by baseline dominant archetype; the biological corner is worst (27%) and the low-burden pole best (60%), a transdiagnostic 27%–60% gradient. (**b**) Held-out incremental value for functioning: the continuous archetype encoding dominates the eight specific axes, any hard tessellation, and the isolated biology axes (operative number of clusters = none). (**c**) The map and DSM-5 are co-informative—each adds beyond the other, and both together add most—but the individual-level remission AUC gain is only +0.011.

receiving (attenuation 4.7%; 3.9% under attrition weighting), so the biological phenotype is not a treatment proxy. Second, even without *selecting* a drug, the map *describes* who faces a difficult course: a patient's baseline pull toward the biological corner stratified two-year treatment course, with that corner carrying roughly twice the treatment-resistance and side-effect burden of the low-burden corner (resistance 43% vs 20%; significant side-effects 23% vs 11%), beyond baseline severity, substance comorbidity and demographics, and within cohort (Extended Data Fig. F6b). As in the prognostic analysis, this is a group-level monitoring signal, not an individual classifier (the steepest gradient, treatment-resistance, was AUC-marginal under a permutation null). *Clinically,* demonstrating treatment *selection* from this map would require randomized or trial-arm data, which this baseline observational cohort—patients receiving usual care in French expert centres—does not contain.

3 Discussion

Across three cohorts of severe mental illness, a single missingness-aware Bayesian model placed clinical and biological measurements in one transdiagnostic coordinate system and yielded a nine-dimension atlas whose most consequential property is a dissociation: metabolic and inflammatory load are the domains least entangled with overall functional severity. If biological burden simply tracked how impaired a patient is, it would add little to a clinician's global impression. The data say the opposite—biological risk rides on axes the severity picture does not see—and the four archetypal extremes make the consequence concrete. The biological corner and the severe, non-biological corner carry comparable functional burden but opposite biological profiles, and they diverge in prognosis (two-year functional remission 27% vs 32%, the biological corner worse). This is the precise property a biology-aware stratification can exploit: it separates patients who look equally ill but are biologically opposite, and that separation is durable and prognostically meaningful.

This headline is corroborated, in direction, by a large and independent literature, which is why we state it as relative rather than absolute independence. Metabolic burden in severe mental illness is driven by medication, adiposity, age and chronicity more than by diagnosis or symptom severity, is present before chronic severity accrues, and is comparable across schizophrenia, bipolar disorder and depression [17–20, 28]; inflammation marks a subgroup and specific symptoms rather than the severity continuum [29–31]; and the immuno-metabolic-depression programme has independently arrived at a separable biological dimension mapping onto atypical, energy-related symptoms [12–16]. That a completely different method, in different cohorts, converges on a separable biology axis is strong convergent validity. We also state the boundary conditions plainly: case–control inflammatory elevations are real [32–34], a modest severity or duration dose–response for metabolic burden exists [35, 36], metabolic dysfunction does track cognition and functioning in mood disorders [37], and much of the inflammation–symptom association is carried by adiposity rather than inflammation *per se* [38, 39]. Our 0.07–0.12 correlations are best read as a quantification of a known qualitative pattern, sitting at its low end because G indexes *functional burden*—from which biology is most decoupled—rather than the case–control contrast of being ill. That they survive, and even fall slightly under, adjustment for medication, adiposity and site is the strongest available evidence that the dissociation is not a confound.

The second contribution is structural. On these clinical–biological coordinates the patient space is a graded continuum, not a set of discrete biotypes, and it cuts almost entirely across DSM-5. This aligns with normative-modelling work reporting extensive individual heterogeneity in the same disorders [9–11], and it stands in apparent tension with the brain-based B-SNIP biotypes [21, 22]. The tension is more apparent than real: the two programmes interrogate different measurement spaces, B-SNIP itself found DSM diagnoses to be a severity continuum, and recent syntheses endorse both categorical and dimensional views [23]; clustering choices can also resurface apparent biotypes from continuous data [24]. We therefore claim no biotypes *in this nine-dimensional clinical–biological space*—a claim we make rigorous with a single-Gaussian falsification null rather than by the absence of an elbow—not that biotypes are absent from all spaces. Anchoring G on functioning alone also sidesteps the well-known instability in what a symptom-only general factor means [6–8], while remaining consistent with the impairment reading of the p factor [4, 5].

The remaining contributions are deliberately calibrated. The map persists over two years—durable biology, moving symptoms—which both validates the measurement longitudinally and motivates a clinical division of labour between traits to stratify on and states to monitor. It predicts future functioning incrementally beyond diagnosis and severity, but the gain is modest and group-level, it is co-informative with rather than superior to DSM-5, and it is carried by the biological *phenotype* (the archetype corner) rather than by any isolated laboratory axis. We report this last point as an honest, model-specific refinement of our own earlier analyses: on the better-calibrated map, "biology predicts functioning" is a phenotype-level claim. That the nine-dimension map is nonetheless a *sufficient* representation of 143 raw indicators for predicting deterioration, and near-sufficient for recovery, shows the parsimony is faithful rather than lossy. And on observational, expert-centre treatment-as-usual the map does not reliably moderate response: where identification is clean the result is a well-identified null, and where it is not (channelled prescribing) the effect is non-estimable. We regard reporting these modest and null findings as a feature. The field has repeatedly produced biotype and biomarker claims that fail to replicate; a transdiagnostic biology atlas that is explicit about the difference between scientific validity (demonstrated here) and individual-level clinical utility (not demonstrated here) is a useful corrective.

In sum, this atlas reframes a clinically actionable hypothesis with measured coordinates: not "which biotype is this patient?" but "where does this patient sit on continuous, partly biological axes that diagnosis and severity do not capture, and how durable is that position?" The biological axes a patient keeps over time forecast the functional state they move toward—a group-level, transdiagnostic signal that complements diagnosis, identifies the biological corner as a concrete enrichment and monitoring target, and points to a specific, testable metabolic/inflammatory × treatment hypothesis for prospective study. Separating biological burden from clinical severity, and showing that the biological part is stable and prognostic, is the step that makes mechanistic stratification in severe mental illness a measurable rather than a rhetorical goal.

Limitations

Four classes of limitation bound these claims. *Internal validity only:* the map is discovered and validated within FACE, with no external-cohort replication, and several invariance partials are documented rather than hidden (G re-anchors in schizophrenia, which lacks the FAST functioning anchor; the inflammatory axis re-weights across the depression cohort; self-rated mania floors in depression). *Outcomes are scale trajectories, not recorded events:* prognostic endpoints are state transitions defined from repeated functioning and severity scales, not registered relapses or hospitalizations—a deliberate de-scope from event-level prognosis. *Observational treatment:* confounding by indication is the dominant threat (E-values 1.1–1.8), exposures are drug-class rather than protocol, channelled treatments are non-estimable, and the cohort is patients in usual care at expert centres rather than a randomized sample; the treatment analysis can therefore only *bound* moderation, not establish selection. *Measurement coverage and horizon:* two axes are thin (mania has two indicators and is reported as data-limited; substance use is a two-cohort axis), education is 40.8% missing overall (and 74.5% in schizophrenia)—a structured missingness the observed-likelihood design handles but that limits that covariate—developmental risk scores as “state” partly because of recall noise in childhood-adversity reports rather than true change, attrition is informative (addressed by inverse-probability weighting), and the follow-up window is two years across three visits. Establishing individual-level clinical utility or treatment guidance would require incident-event outcomes, randomized or trial-arm treatment data, and external validation beyond this baseline observational cohort.

4 Methods

This section gives a high-level account of the model and the analysis. The complete mathematical development—the generative model, the observed-data likelihood, the rotation and identification argument, the closed-form covariance identity, the equivalence between the marginalized Bayesian model and full-information maximum likelihood, and the stratification, temporal, prognostic and treatment estimators—is given with proofs in the Annexes (A–E).

4.1 Cohorts, harmonization and the no-imputation data layer

Data were the baseline (visit V0) of three FACE cohorts (Fondation FondaMental expert centres): bipolar disorder, schizophrenia and major depression (N = 9,013; BP 6,252, SZ 2,209, DR 552; 21 recruitment sites). A harmonized data dictionary maps each modelled variable to its per-cohort source column, a harmonization rule and per-variable plausibility (“sanity”) bounds; out-of-range values are set to missing (never clipped, never imputed). Deterministic skip-logic decoding sets structural zeros only where a gate is explicitly negative (for example, attempt count = 0 when ever-attempted = 0); this is decoding, not imputation. Each variable carries a likelihood family and a missingness type. Identifiers (patient, cohort, DSM-5 subtype, visit, site) are never modelled as indicators; cohort and subtype enter only as covariates, invariance-grouping variables or validation labels. Of the 201 usable harmonized common variables, 143 entered as modelled indicators; the remaining 58 are identifiers, covariates or validation labels and are never factor indicators.

4.2 Candidate ontology and soft priors

Ten candidate constructs defined a *soft-prior* ontology rather than a fixed output. Mapping the candidates against the common-variable coverage fixes the eligible set and is itself a primary result: a construct with no common indicator is reported *not testable*, never back-filled with an invented proxy. Each (indicator, factor) cell receives a prior *role*—primary, plausible-cross, unlikely or forbidden—whose default is shrinkage rather than exclusion, so the model can confirm, split, merge, downgrade, reject or declare not-testable any candidate. The formal prior is given in Annex A.

4.3 The measurement model

The measurement model rests on five core equations; their full derivation, with proofs, is in Annex A. We posit that a few latent factors generate the many measured quantities, so each patient is summarized by a low-dimensional coordinate. Patient i carries a factor vector of one general functional-burden axis G and $K = 8$ specific axes, given a standardized multivariate-normal prior,

$$f_i = (G_i, D_{i1}, \dots, D_{iK})^\top \;\sim\; \mathcal{N}_{K+1}(\mathbf{0}, \Phi)\,, \qquad \operatorname{diag}(\Phi) = \mathbf{1}. \tag{4}$$

The unit diagonal of the factor correlation matrix Φ fixes the latent scale; G is the burden axis every indicator may share, while $D_{i1}, \dots, D_{iK}$ are the domain-specific axes (cognition, metabolic, inflammatory, sleep, developmental risk, suicidality, mania and substance use). Equation (4) is the formal content of the idea that a patient is *one general burden axis plus specific axes.*

Each indicator j is a linear reflection of those factors through a single predictor,

$$\eta_{ij} = \alpha_j + \lambda_{jG}\, G_i + \sum_{k=1}^{K} \lambda_{jk}\, D_{ik} + \beta_j^\top c_i, \tag{5}$$

with item intercept α_j, a general loading λ_{jG} onto G, specific loadings λ_{jk}, and item-level covariates c_i that shift an indicator's mean but are never themselves factor indicators. Equation (5) *is* the bifactor exploratory structural-equation model: bifactor because every indicator may load on G *and* on its home specific factor; exploratory because small cross-loadings λ_{jk} are allowed but shrunk toward zero by soft priors rather than forbidden outright. The predictor feeds a per-type observation density—Gaussian or log-Gaussian for continuous and laboratory measures, ordered-logit for ordinal, Bernoulli for binary, negative-binomial for counts—and continuous indicators additionally enter through a rank-based *Gaussian-copula* marginal, so that forcing heterogeneous variables onto a shared scale never manufactures covariance. The likelihood is invariant to rotation of the factors; identification comes from the soft loading priors, so a prior-free refit that recovers the same structure shows the map is earned from the data, not imposed (Annex A).

Given the latent scores the indicators are conditionally independent (local independence: once the factors are fixed, only item-specific noise remains, a diagonal residual covariance Ψ). Equations (4)–(5) then generate a structured covariance among the indicators,

$$\Sigma = \Lambda\,\Phi\,\Lambda^\top + \Psi = \underbrace{\lambda_G \lambda_G^\top}_{\text{general: rank one}} + \underbrace{\Lambda_D \Phi_D \Lambda_D^\top}_{\text{specific factors}} + \underbrace{\Psi}_{\text{private noise}}, \tag{6}$$

where $\Lambda = [\,\lambda_G \mid \Lambda_D\,]$ collects the loadings and Φ_D is the specific-factor correlation block. Equation (6) is the generative claim in one line—G leaves a rank-one footprint that touches every item, the specific factors contribute a block, and Ψ absorbs what no factor explains—and it is also why imputation is forbidden: a filled-in cell would inject covariance that (6) would misread as a factor.

Accordingly the model is fit to each patient's *observed cells only.* With O_i the set of indicators observed for patient i and F_j the density of indicator j, the observed-data log-likelihood is

$$\log p(X_{\text{obs}} \mid \Theta) = \sum_{i=1}^{N} \sum_{j \in O_i} \log F_j(X_{ij} \mid \eta_{ij}, \theta_j)\,. \tag{7}$$

A missing cell contributes no term and no completed $N \times J$ matrix is ever formed; deterministic skip-logic decodes structural zeros as *observed* values, not imputed ones. For the multivariate-normal block this observed-cell sum is exactly the full-information maximum-likelihood objective for incomplete data, obtained by *marginalizing* the missing coordinates rather than filling them (Annex A). Equation (7) is the no-imputation invariant on which every downstream analysis rests.

Two factor covariances are estimated. The primary *bifactor* fit holds G orthogonal to the specific axes (Φ block-diagonal); this is the natural null for the biology question. A *correlated-G* arm then unzips that zero block,

$$\Phi = \begin{pmatrix} 1 & \phi_{Gd}^{\top} \\ \phi_{Gd} & \Phi_D \end{pmatrix}, \qquad \phi_{Gd} = \operatorname{Corr}(G, D) \in [-1, 1]^K, \tag{8}$$

turning "biology is separable from severity" from a built-in constraint into the *measured* vector ϕ_{Gd}: a domain is severity-entangled when its entry is large and separable when it is near zero (metabolic 0.12 and inflammatory 0.07, versus cognition 0.39 and sleep 0.42). Reporting the bifactor general loadings $|\lambda_{jG}|$ and the correlated-G correlations ϕ_{Gd} together—they agree on the ordering—is what makes the separation a measurable estimand rather than an artefact of the orthogonality constraint (Figure 4; Annex A). The biology–G result is reported unadjusted and under a covariate-adjustment ladder (age, sex, education, site, antipsychotic exposure, adiposity) implemented by residualizing each indicator before the factor model.

4.4 Estimation, confirmation, invariance and scoring

The global posterior was reached by continuation: a well-conditioned sub-model was fit and then deformed toward the full target one source of difficulty at a time, each converged posterior warm-starting the next, and *only the global fit is interpreted.* The Gaussian block is marginalized analytically (Woodbury identity and matrix-determinant lemma, grouped by missingness pattern), which removes the per-patient latent funnel and makes full-sample estimation tractable on a workstation (Annex A). Sampling used the No-U-Turn sampler [40] in PyMC [41] with the NumPyro/JAX backend [42]; a stage advanced only on standard convergence criteria ($\hat{R} \leq 1.01$, adequate effective sample size, zero divergences, no Heywood cases, acceptable posterior-predictive behaviour), with a cross-seed stability guard for the joint mixed fit. Confirmation was performed in-engine: a flat-prior refit (identification constraints only) reproduced the loadings and factor correlations (Tucker congruence $\varphi = 1.00$); posterior-predictive checks gave a Bayesian standardized root-mean-square residual ≈ 0.07 for the continuous block and reproduced 21/22 non-Gaussian indicators within the 90% predictive interval; and model comparison by WAIC and PSIS-LOO [43, 44] preferred the bifactor over unidimensional and correlated-factor alternatives. Because the marginalized Bayesian model and full-information maximum likelihood optimize the same observed-data objective (Annex A), a standalone maximum-likelihood arm adds no independent evidence. Measurement invariance across BP/SZ/DR was tested by multi-group loadings and Bayesian differential item functioning, and robustness by leave-one-cohort-out congruence, diagnosis-balanced subsampling, site cluster-bootstrap and inverse-cohort weighting (minimum Tucker $\varphi = 0.96$). Each patient received, per axis, a posterior mean, standard deviation and credible interval, the number of observed indicators and a reliability tier, so that all downstream analyses propagate uncertainty rather than treating coordinates as equally measured.

4.5 Stratification, temporal coherence, prognosis and treatment

On the fitted coordinates $\{x_i, S_i\}$—each patient's posterior-mean position $x_i = \hat{f}_i \in \mathbb{R}^9$ together with its posterior covariance S_i—structure was tested *before* any clustering by a multi-statistic battery computed over the posterior draws, anchored by a *single-Gaussian falsification null*: synthetic data are drawn from one Gaussian with the empirical mean and covariance of the coordinates (structureless by construction), the identical pipeline is run on real and null data, and we ask whether the real cloud's best partition separates patients any better than the null's. It does not (best-silhouette $z = -0.36$; Annex B), so the verdict is a graded *continuum*, not discrete biotypes. The load-bearing representations are therefore the continuous coordinates themselves and a soft archetype simplex, not any hard partition. Archetypal analysis [45] writes each patient as a convex blend of $A = 4$ extreme phenotypes pinned to the convex hull of the cloud (the barycentric representation reported in Eq. 3), each archetype itself a convex combination of patients; the simplex weights $w_i \in \Delta^{A-1}$ are a patient's soft archetype weighting, so the continuum is *read* through extremes rather than *cut* into kinds. The number of corners ($A = 4$) was set by cross-seed stability rather than parsimony, with the full optimization $z_a = \sum_j b_{ja}\hat{f}_j$ and the draw-wise uncertainty propagation given in Annex B. A measurement-error Gaussian mixture (extreme deconvolution [46], each patient carrying its own S_i) supplies a nested family of soft tessellations with no privileged number of regions, the operative granularity deferred to incremental predictive validity downstream (Annex B). Each downstream analysis then treats the measurement model as *fixed*—reusing or re-scoring coordinates without re-discovering the map—and propagates each coordinate's posterior uncertainty S_i rather than treating the scores as exact. Temporal coherence scored the follow-up visits on the *fixed* measurement model (observed cells, uncertainty propagated), tested longitudinal invariance, and decomposed trait from state with a measurement-error random-intercept model that plugs the known baseline measurement variance and includes visit fixed effects, yielding an individual rank-stability intraclass correlation and a population-level shift (Annex C). Prognosis compared nested models—nuisance plus baseline outcome plus severity, then adding DSM-5 subtype, the map, or both—using an errors-in-variables Bayesian generalized linear model that propagates each coordinate's posterior standard deviation, scored by held-out expected log predictive density and, for binary endpoints, by change in AUC; robustness used inverse-probability weighting, restriction to reliable axes and a permutation null, and a separate benchmark compared the map against all 143 raw indicators under a matched gradient-boosting classifier (Annex D). Treatment exposures, harmonized to drug classes, were analysed by an identification-first causal pipeline—common-support overlap gate, stabilized inverse-probability-of-treatment weighting on a propensity that includes the nine coordinates [47, 48], doubly-robust errors-in-variables moderation, an E-value sensitivity analysis [49] and a minimum-detectable-effect calculation that makes any null a *bounded* one (Annex E).

4.6 Compute and reproducibility

The pipeline is configuration-first with fixed seeds. Raw per-cohort data are confidential and never leave their governed environment, while the code, the harmonized dictionary and the regenerated aggregate results are shareable; every reported number is reproducible from the numbered pipeline, and the sample-characteristics table is regenerated by an aggregate-only script. Estimation ran on a workstation (Apple M4 Pro, 24 GB; no GPU).

Data availability

The FACE database is the property of Fondation FondaMental; per-patient cohort data are confidential and governed by FondaMental's data-access procedures, to which qualified researchers may apply. No per-patient data are reproduced in this article. Aggregate parameters, derived structure and all figure/table source values are available in the project's regenerated aggregate outputs.

Code availability

The full analysis pipeline (data harmonization, the Bayesian measurement engine, stratification, temporal, prognosis and treatment modules), the harmonized variable dictionary, and configuration files are available at [**TODO:** repository URL / DOI]. Every reported quantity is reproducible from the numbered pipeline; per-patient data are never required to regenerate the aggregate results, tables and figures.

Ethics

The FACE cohorts were approved by the relevant institutional review boards and all participants provided written informed consent. [**TODO:** Insert exact IRB/ethics committee names, approval numbers, and trial/cohort registration identifiers.]

Author contributions

[**TODO:** Complete per ICMJE / journal policy and FACE–FondaMental authorship conventions, including cohort investigators.]

Funding

[**TODO:** Fondation FondaMental and cohort funders to be specified.]

Competing interests

[**TODO:** Declare per journal policy.]

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Annexes — full derivations, proofs and secondary materials

A The measurement model: full derivation

This annex develops the global measurement model from first principles: the generative factor model and its defining local-independence assumption (A.1); the mixed-likelihood observation model and the Gaussian-copula marginal (A.2); the observed-data likelihood and why missingness is handled by marginalization rather than imputation (A.3); the soft-prior loadings and the rotation problem they resolve (A.4); the closed-form covariance identity that carries the headline estimand (A.5); the Woodbury evaluation and its exact equivalence with full-information maximum likelihood (A.6); and the staged estimation and in-engine confirmation (A.7). Propositions are proved; everything reported in the main text is an evaluation of the objects defined here.

A.1 Generative model and local independence

Index patients $i = 1, \dots, N$, indicators $j = 1, \dots, J$, specific factors $k = 1, \dots, K$ and one general factor G. A few latent causes generate the many measured quantities: each patient carries a factor vector $f_i = (G_i, D_{i1}, \dots, D_{iK})^\top \in \mathbb{R}^{K+1}$ with

$$f_i \;\sim\; \mathcal{N}_{K+1}(\mathbf{0},\, \Phi)\,, \qquad \operatorname{diag}(\Phi) = \mathbf{1}, \tag{A9}$$

the factor *correlation* matrix Φ carrying a unit diagonal to fix the latent scale. Each indicator is a linear reflection of the factors plus item-specific noise, through a single linear predictor

$$\eta_{ij} \;=\; \alpha_j \;+\; \lambda_{jG}\, G_i \;+\; \sum_{k=1}^{K} \lambda_{jk}\, D_{ik} \;+\; \beta_j^\top c_i, \tag{A10}$$

with item intercept α_j, general loading λ_{jG}, specific loadings λ_{jk} (row j of the loading matrix $\Lambda = [\,\lambda_G \mid \Lambda_D\,] \in \mathbb{R}^{J\times(K+1)}$), and item-level covariates c_i that adjust an indicator's *mean* but are never factor indicators. Stacking one patient's items, with the general column separated from the specific block,

$$x_i = \mu_i + \underbrace{\lambda_G G_i}_{\text{general}} + \underbrace{\Lambda_D D_i}_{\text{specifics}} + \varepsilon_i, \qquad \mu_i = \alpha + B c_i, \qquad \varepsilon_i \sim \mathcal{N}(\mathbf{0}, \Psi), \tag{A11}$$

with diagonal residual covariance $\Psi = \operatorname{diag}(\sigma_1^2, \dots, \sigma_J^2)$.

Local independence (the defining assumption). Given a patient's latent scores, the items are conditionally independent,

$$p(x_i \mid f_i, \Theta) = \prod_{j\in O_i} p(x_{ij} \mid f_i, \Theta), \tag{A12}$$

i.e. the factors are the only reason items covary. In the Gaussian linear model this is *identical* to "Ψ is diagonal": local independence and diagonal residual covariance are two faces of one assumption—that the factors explain away all shared covariance. (It is also why filling missing cells is forbidden: an invented value injects covariance the model would mistake for a factor.) Because f_i is unobserved and the likelihood must depend on Θ alone, we integrate it out and multiply across independent patients,

$$L(\Theta) = \prod_{i=1}^{N} p(x_i \mid \Theta) = \prod_{i=1}^{N} \int \Big[\prod_{j\in O_i} p(x_{ij} \mid f_i, \Theta)\Big]\, \mathcal{N}(f_i; \mathbf{0}, \Phi)\, \mathrm{d}f_i. \tag{A13}$$

This is the complete first-principles likelihood: a product over patients (independent units), a product over items (local independence), and an integral that removes the latent (the sum rule). Equation (A10) is a bifactor exploratory structural-equation model [25, 26]—bifactor because every indicator may load on G *and* on its home specific factor; exploratory because an indicator is *allowed* small cross-loadings on plausibly related factors, shrunk toward zero by the soft priors (A.4), rather than forbidden outright as in confirmatory factor analysis.

A.2 Mixed-likelihood observation model and the Gaussian copula

Coercing variables onto a common scale would manufacture covariance and bias the very factors being estimated. Each indicator instead keeps the observation density appropriate to its type, all sharing the linear predictor (A10):

$$\text{continuous / } z\text{-score:} \quad X_{ij} \sim \mathcal{N}(\eta_{ij},\ \sigma_j^2), \tag{A14}$$

$$\text{skewed laboratory:} \quad \log X_{ij} \sim \mathcal{N}(\eta_{ij},\ \sigma_j^2), \tag{A15}$$

$$\text{ordinal } (C_j \text{ levels}): \quad \Pr(X_{ij} \le c) = \operatorname{logit}^{-1}(\tau_{jc} - \eta_{ij}), \quad \tau_{j1} < \cdots < \tau_{j,C_j-1}, \tag{A16}$$

$$\text{binary:} \quad X_{ij} \sim \operatorname{Bernoulli}\Big(\operatorname{logit}^{-1}(\eta_{ij})\Big), \tag{A17}$$

$$\text{count:} \quad X_{ij} \sim \operatorname{NegBinom}(\mu_{ij} = e^{\eta_{ij}},\ \phi_j)\,. \tag{A18}$$

The Gaussian-copula marginal (reported map). The reported map replaces the raw Gaussian likelihood on the continuous block with a semiparametric *Gaussian-copula* model: each continuous indicator j is mapped to a latent Gaussian score by its frozen empirical distribution function,

$$z_{ij} = \Phi_{\mathcal{N}}^{-1}(\hat{F}_j(X_{ij})), \tag{A19}$$

and the factor model of A.1 is fit on z_{ij}. This makes the continuous block invariant to monotone re-scaling and robust to non-normal margins (heavy tails, skew, bounded scales) while leaving the latent dependence structure—the loadings and factor correlations that the scientific claims live in—unchanged. Because the rank transform (A19) is monotone, it alters none of the identification arguments below. Fitting the same structure under the plain Gaussian likelihood recovers the same loadings and correlations to within seed variability; that the central findings hold under both is reported in the main text as a robustness result. Writing $F_j(\cdot)$ for indicator j's density and $O_i \subseteq \{1, \dots, J\}$ for its *observed* indicators, the observed-data log-likelihood sums over observed cells only,

$$\log p(X_{\text{obs}} \mid \Theta) = \sum_{i=1}^{N} \sum_{j \in O_i} \log F_j(X_{ij} \mid \eta_{ij}, \theta_j)\,, \tag{A20}$$

so a missing cell contributes no term and no completed $N \times J$ matrix is ever formed; deterministic skip-logic decodes structural zeros as *observed* values.

A.3 Missing data is the marginal property, not imputation

Proposition 1 (Marginalization)**.** *For patient i with observed continuous set $C_i \subseteq O_i$, the model-implied distribution of the observed sub-vector is obtained by* deleting *the unobserved coordinates from (μ, Σ):*

$$X_{i,C_i} \sim \mathcal{N}(\mu_{i,C_i},\ \Sigma_{C_i})\,, \qquad \Sigma_{C_i} = \Lambda_{C_i} \Phi \Lambda_{C_i}^{\top} + \Psi_{C_i}, \tag{A21}$$

with Λ_{C_i} the observed-row loadings and $\Psi_{C_i} = \operatorname{diag}(\sigma_j^2)_{j \in C_i}$.

Proof. A foundational property of the multivariate normal is that marginals are read off by deleting coordinates: if $x \sim \mathcal{N}(\mu, \Sigma)$ then any sub-vector x_C is $\mathcal{N}(\mu_C, \Sigma_{CC})$. Apply this to $x_i \sim \mathcal{N}(\mu_i, \Sigma)$ with $\Sigma = \Lambda\Phi\Lambda^\top + \Psi$ (Proposition 3); the submatrix Σ_{CC} retains exactly the observed rows/columns of Λ and the observed diagonal of Ψ, giving (A21). □

Hence "integrating out" the missing cells is not an extra step but the marginal property itself; summing $\log \mathcal{N}(\cdot)$ over each patient's observed sub-vector is precisely the full-information maximum-likelihood (FIML) objective for incomplete multivariate-normal data. This is the formal justification of the no-imputation invariant: any completed matrix would inject between-cell covariance that (A12) would misattribute to a latent factor.

A.4 Soft-prior loadings, identification and the rotation problem

Theory enters through the prior on each loading. Each (indicator, factor) cell is assigned a role $\rho(j,k)$—one of PRIMARY, plausible CROSS, UNLIKELY, or FORBIDDEN—and the prior on that loading is

$$\lambda_{jk} \mid \rho(j,k) \;\sim\; \begin{cases} \mathcal{N}_+(0.70,\, 0.25^2) & \text{PRIMARY (HOME CELL)}, \\ \mathcal{N}(0,\, 0.25^2) & \text{PLAUSIBLE CROSS}, \\ \mathcal{N}(0,\, 0.05^2) & \text{UNLIKELY}, \\ \delta_0 & \text{FORBIDDEN}, \end{cases} \tag{A22}$$

where $\mathcal{N}_+$ truncates to $(0, \infty)$ and δ_0 fixes the loading at zero. Remaining priors are weakly informative: $\alpha_j \sim \mathcal{N}(0, 1.5^2)$; $\sigma_j = 0.05 + \mathrm{HalfNormal}(1)$ (the floor guards against a Heywood case); ordered-logit cutpoints $\tau_{j\cdot}$ ordered; the negative-binomial concentration half-normal; and $\Phi \sim \mathrm{LKJ}(\eta{=}2)$, whose density $\propto \det(\Phi)^{\eta-1}$ gently favours weakly correlated factors [50]. Latent scores use a non-centred parameterization to defuse the hierarchical funnel.

Proposition 2 (Rotation invariance of the likelihood)**.** *For any invertible $M \in \mathbb{R}^{(K+1)\times(K+1)}$, the reparameterization $\Lambda^\star = \Lambda M$, $\Phi^\star = M^{-1}\Phi M^{-\top}$, $f^\star = M^{-1} f$ leaves the implied covariance, and hence the Gaussian likelihood, unchanged:*

$$\Lambda^\star\Phi^\star\Lambda^{\star\top} + \Psi = \Lambda M\,(M^{-1}\Phi M^{-\top})\,M^\top\Lambda^\top + \Psi = \Lambda\Phi\Lambda^\top + \Psi = \Sigma. \tag{A23}$$

Proof. Immediate from $MM^{-1} = I$: the inner $M(M^{-1}\Phi M^{-\top})M^\top$ telescopes to Φ. Since x_i is determined in distribution by (μ_i, Σ) and Σ is invariant, so is the likelihood. □

A factor model is therefore not identified from the likelihood alone; confirmatory factor analysis breaks this by fixing loadings to exactly zero, whereas the soft-prior model breaks it with the prior. The likelihood is rotation-invariant but the prior is *not*: a rotation moves a loading the prior wanted near zero away from zero, lowering $p(\Lambda)$, so the posterior $p(\Lambda \mid X) \propto p(X \mid \Lambda)\, p(\Lambda)$ concentrates at the orientation where small loadings align with the soft-zero priors and home loadings are positive and substantial; the positive truncation removes the residual sign-flip.

Corollary 1 (Flat-prior confirmation)**.** *Under a flat loading prior (identification constraints only) the posterior mode equals the maximum-likelihood (FIML) solution. Hence a prior-free refit landing on the same loadings demonstrates that the structure is earned from the data, not manufactured by the priors. Empirically this refit reproduced the map to Tucker congruence $\varphi = 1.00$.*

A.5 The covariance identity and the headline estimand

Proposition 3 (Bifactor covariance)**.** *Under* (A11) *with* $\mathrm{Var}(G_i) = 1$, $\mathrm{Cov}(D_i) = \Phi_D$, $\mathbb{E}[\varepsilon_i\varepsilon_i^\top] = \Psi$ *diagonal, and the bifactor orthogonality* $\mathbb{E}[G_i D_i^\top] = \mathbf{0}$ *together with* $(G_i, D_i) \perp \varepsilon_i$*,*

$$\Sigma = \mathrm{Cov}(x_i) = \underbrace{\lambda_G\lambda_G^\top}_{\textit{general: rank one}} + \underbrace{\Lambda_D\Phi_D\Lambda_D^\top}_{\textit{specific factors}} + \underbrace{\Psi}_{\textit{private noise}} = \Lambda\,\Phi\,\Lambda^\top + \Psi, \qquad \Phi = \begin{pmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & \Phi_D \end{pmatrix}. \tag{A24}$$

Proof. Expand $\Sigma = \mathbb{E}[(x_i - \mu_i)(x_i - \mu_i)^\top]$ with $x_i - \mu_i = \lambda_G G_i + \Lambda_D D_i + \varepsilon_i$. The nine resulting terms include cross-terms $\lambda_G\,\mathbb{E}[G_i D_i^\top]\,\Lambda_D^\top$ and $\mathbb{E}[(\lambda_G G_i + \Lambda_D D_i)\varepsilon_i^\top]$; the first vanishes by $\mathbb{E}[G_i D_i^\top] = \mathbf{0}$ and the second by independence of the factors and the noise. The survivors are $\lambda_G\lambda_G^\top\,\mathrm{Var}(G_i)$, $\Lambda_D\Phi_D\Lambda_D^\top$ and Ψ, giving (A24); the zero off-diagonal block of Φ encodes $G \perp$ specifics. $\square$

The general factor leaves a *rank-one* footprint: every pair of items covaries through G by exactly $\lambda_{jG}\lambda_{j'G}$—one shared source touching all items.

The headline lives in the off-diagonal block. The correlated-G model *unzips* the zero block, freeing $\phi_{Gd} = \mathrm{Corr}(G, D)$:

$$\Phi = \begin{pmatrix} 1 & \phi_{Gd}^\top \\ \phi_{Gd} & \Phi_D \end{pmatrix}, \qquad \Sigma = \lambda_G\lambda_G^\top + \underbrace{\lambda_G\phi_{Gd}^\top\Lambda_D^\top + \Lambda_D\phi_{Gd}\lambda_G^\top}_{\text{the terms orthogonality removed}} + \Lambda_D\Phi_D\Lambda_D^\top + \Psi. \tag{A25}$$

The vector ϕ_{Gd}—the correlation of each specific factor with general burden—is the headline estimand of the enterprise: "biology is the least severity-entangled domain" is the statement that its metabolic and inflammatory entries are near zero (0.12 and 0.07) while cognition and sleep are far larger (0.39 and 0.42). The bifactor orthogonality is the *null*; the correlated-G arm measures the departure from it; and the bifactor direct loading $|\lambda_{jG}|$ and the correlation ϕ_{Gd} agree on the ordering—reporting both is what turns "biology is separable" from a constraint artefact into a finding.

A.6 Woodbury evaluation and equivalence with FIML

Evaluating (A21) naively costs $\mathcal{O}(|C_i|^3)$ per patient. The Woodbury identity and matrix-determinant lemma reduce it to the $(K{+}1)$-dimensional factor space.

Lemma 1 (Woodbury reduction)**.** *With* Ψ_{C_i} *diagonal and* Λ_{C_i} *the observed-row loadings,*

$$\Sigma_{C_i}^{-1} = \Psi_{C_i}^{-1} - \Psi_{C_i}^{-1}\Lambda_{C_i}\big(\Phi^{-1} + \Lambda_{C_i}^\top\Psi_{C_i}^{-1}\Lambda_{C_i}\big)^{-1}\Lambda_{C_i}^\top\Psi_{C_i}^{-1}, \tag{A26}$$

$$\log|\Sigma_{C_i}| = \log\left|\Phi^{-1} + \Lambda_{C_i}^\top\Psi_{C_i}^{-1}\Lambda_{C_i}\right| + \log|\Phi| + \sum_{j\in C_i}\log\sigma_j^2. \tag{A27}$$

The inner solve is $\mathcal{O}((K{+}1)^3)$, independent of $|C_i|$; patients sharing a missingness pattern share the inner $(K{+}1)\times(K{+}1)$ factorization, computed once per pattern. This removes the per-patient latent funnel for the continuous block and makes full-sample ($N = 9{,}013$) estimation tractable on a workstation; the non-Gaussian block retains explicit per-patient latents.

Proposition 4 (Bayes/FIML equivalence)**.** *The marginalized Bayesian observed-data model and FIML optimize the* same *observed-data likelihood and differ only by the priors* (A22)*. In particular (Corollary 1) the flat-prior MAP equals the FIML estimate, so a standalone FIML estimator adds no independent evidence and confirmation is done in-engine.*

Proof. By Proposition 1 the per-patient marginal density is the incomplete-data multivariate-normal density on C_i, whose log summed over patients is the FIML objective. The posterior is this likelihood times the prior; with a flat prior the log-posterior equals the log-likelihood up to a constant, so its mode is the FIML maximizer. $\square$

Factor scores. Each patient's coordinates are the expected a posteriori (EAP) factor scores from their observed cells; for an all-continuous pattern,

$$\hat{f}_i = \mathbb{E}[f_i \mid X_{i,C_i}] = \Phi\Lambda_{C_i}^{\top}\Sigma_{C_i}^{-1}(X_{i,C_i} - \mu_{i,C_i}), \quad S_i = \Phi - \Phi\Lambda_{C_i}^{\top}\Sigma_{C_i}^{-1}\Lambda_{C_i}\Phi, \tag{A28}$$

obtained by MCMC when non-Gaussian cells are present. The per-patient posterior covariance S_i is exported and propagated downstream (it is the noise term in the stratification mixture, Annex B): a patient with one observed indicator for a factor is *not* treated as equally characterized as one with six.

A.7 Staged estimation and in-engine confirmation

The global posterior was reached by continuation (homotopy): a well-conditioned sub-model was fit and then deformed toward the full target one source of difficulty at a time (continuous core at full N; exploratory cross-loadings; mixed-likelihood suicidality and developmental risk; thin cohort-specific factors; all factors jointly plus the correlated-G variant), each converged posterior warm-starting the next. *Only the global fit is interpreted*; the earlier stages are convergence/identification checkpoints. Sampling used the No-U-Turn sampler [40] via PyMC [41] with the NumPyro/JAX backend [42]; a stage advanced only on $\hat{R} \leq 1.01$, effective sample size ≥ 400, zero divergences, no Heywood cases and acceptable posterior-predictive behaviour. The full-N continuous backbone met this strict gate; the joint nine-dimension (mixed) map is reported at the largest sample that mixes ($\hat{R} \leq 1.04$) with a cross-seed stability guard (Tucker congruence $\varphi = 0.99$). Confirmation used (i) the flat-prior refit (Corollary 1); (ii) posterior-predictive checks (continuous Bayesian standardized root-mean-square residual ≈ 0.07; 21/22 non-Gaussian indicators within the 90% predictive interval, the lone exception a 90.8%-zero suicide-attempt count item—an item-level, not factor-level, misfit); and (iii) model comparison by WAIC and PSIS-LOO [43, 44], which decisively preferred the bifactor over unidimensional and correlated-factor alternatives. Classical fit indices (CFI/TLI, RMSEA) are not reported—they are weak under heavy missingness and non-normal indicators and add no independent evidence here.

B Stratification: structure testing, archetypes and soft regions

The stratification operates on the fitted coordinates $\{x_i, S_i\}_{i=1}^{N}$ with $x_i = \hat{f}_i \in \mathbb{R}^D$ ($D = 9$) and known posterior covariance S_i from (A28); uncertainty is carried throughout rather than discarded.

B.1 Structure-discovery gate and the single-Gaussian falsification null

Before any clustering we test the cloud's shape with complementary statistics, each computed over posterior draws so measurement noise cannot manufacture structure. *Cluster tendency* uses the Hopkins statistic $H = \sum_i u_i^D / (\sum_i u_i^D + \sum_i w_i^D)$, where w_i is the nearest-neighbour distance of a sampled real point and u_i that of a uniform point over the data support ($H \approx \frac{1}{2}$ indicates no clustering); *modality* uses Hartigan's dip statistic [51] on PC1 and each axis; *model selection* uses the gap statistic [52] and the Gaussian-mixture BIC profile; *density* uses HDBSCAN [53] (fraction of points assigned to noise); *topology* a Mapper graph [54]; and *separation* the best silhouette over K. A continuum verdict requires $H \to \frac{1}{2}$, a unimodal dip, the gap and BIC failing to prefer $K > 1$, HDBSCAN near-all-noise, and a single Mapper component. Empirically: $H = 0.80$, PC1 dip $p \approx 1.0$, silhouette peak 0.14, HDBSCAN 0 clusters (100% noise), Mapper a single connected component.

Because all of these are absolute statistics, we anchor them with a *falsification null.* We draw synthetic data from a single multivariate Gaussian with the empirical mean and covariance of the real coordinates—a structureless cloud by construction—and run the identical pipeline. If the real cloud were truly clustered, its best partition would separate points *better* than the null's.

$$z_{\mathrm{sil}} = \frac{s^{\star}_{\mathrm{real}} - \mathbb{E}[s^{\star}_{\mathrm{null}}]}{\mathrm{sd}[s^{\star}_{\mathrm{null}}]} = \frac{0.140 - 0.141}{0.002} = -0.36, \tag{B29}$$

i.e. the real data are *not* better separated than a Gaussian blob; the mixture-optimal number of components collapsed to one in all 20 posterior draws. This turns "no elbow" into a calibrated negative result.

B.2 Archetypal analysis (the load-bearing continuum view)

Each patient is represented as a convex combination of A extreme phenotypes, each itself a convex combination of patients [45]:

$$\min_{W,B} \sum_{i=1}^{N} \Big\| x_i - \sum_{a=1}^{A} w_{ia}\, z_a \Big\|^2 \quad \text{s.t.} \quad z_a = \sum_{j=1}^{N} b_{ja}\, x_j, \quad w_{ia}, b_{ja} \geq 0, \quad \textstyle\sum_a w_{ia} = 1, \ \sum_j b_{ja} = 1, \tag{B30}$$

i.e. $X \approx WZ$ with $Z = BX$ and both W and B row-stochastic, which forces the archetypes z_a onto the convex hull of the cloud. The simplex weights $w_i \in \Delta^{A-1}$ are the soft archetype membership. Uncertainty is propagated by refitting (B30) across posterior draws and projecting each draw onto the fixed archetypes (simplex-constrained least squares). The number of corners A is set by *cross-seed stability* (largest A whose minimum cross-seed Tucker congruence ≥ 0.8), not by parsimony: only $A = 2$ (0.999) and $A = 4$ (0.98, minimum congruence 0.997 over three seeds at full N) were stable, while $A = 3, 5, \ldots, 8$ collapsed (0.05–0.51). The operational rule selected $A = 4$ (explained variance 0.52). This is copula-specific: the higher-A corners of a plain-Gaussian map do not reproduce once the honest, wider explicit-axis uncertainty is carried. Because biological load varies continuously without a density gap, it is a *direction of maximal phenotype*—an archetypal corner—rather than a partition boundary, which is why the archetypes (and the continuous coordinates), not any tessellation, are the load-bearing object for biology.

B.3 Measurement-error mixture and the nested K-family

Treating x_i as a noisy reading of a latent position, $x_i \mid \theta_i \sim \mathcal{N}(\theta_i, S_i)$, with θ_i from a K-component mixture yields the extreme-deconvolution likelihood [46]

$$x_i \;\sim\; \sum_{k=1}^{K} \pi_k\, \mathcal{N}(m_k,\ \Sigma_k + S_i)\,, \qquad \pi \sim \mathrm{Dirichlet}\big(\tfrac{\alpha}{K}\mathbf{1}\big)\,, \tag{B31}$$

in which *every* observation carries its own additive, known covariance S_i, so imprecise coordinates spread their mass over components and prior-dominated axes self-down-weight. The soft regions are the posterior responsibilities $r_{ik} = \pi_k \mathcal{N}(x_i \mid m_k, \Sigma_k + S_i) / \sum_l \pi_l \mathcal{N}(x_i \mid m_l, \Sigma_l + S_i)$. On a continuum the mixture BIC is essentially flat (197,900–198,100 across $K = 2$–8), so no K is privileged; we export a nested family ($K = 2, 3, 4$) as labelling conventions and *defer the operative granularity to incremental predictive validity* (Annex D), which returns "no hard K". A coarsest-K default ($K = 2$) is carried only as a hand-off contract.

B.4 Validation gates

The representation is tested on four gates. *Not-just-severity*: variance explained per axis $\eta_d^2 = 1 - \sum_k \sum_{i \in \mathcal{C}_k} (x_{id} - \bar{x}_{kd})^2 / \sum_i (x_{id} - \bar{x}_d)^2$ peaks on the specific axes (mean $\eta^2_{\text{spec}} = 0.12$) far above G (0.008). *Transdiagnosticity*: the adjusted Rand index ARI = (RI − $\mathbb{E}$[RI])/(max RI − $\mathbb{E}$[RI]) against cohort and the seven DSM-5 subtypes is ≈ 0 (0.00 and 0.01). *Stability and not-a-missingness-artefact*: cross-seed ARI 0.998, and a classifier predicting region membership from each patient's per-axis coverage pattern underperforms the majority baseline (lift −0.05, permutation $p = 0.23$). *Tighter than DSM-5*: a free mixture beats one constrained to the DSM-5 partition by information criterion (197,900 vs 201,300). Embeddings (PCA, Figure 5; UMAP [55]) are used for visualization only, never as clustering inputs.

C Temporal coherence: forward scoring, invariance and trait/state

C.1 Forward scoring under the fixed measurement model

Follow-up visits were scored on the *fixed* measurement model: the baseline ($V0$) parameters $(\hat{\Lambda}, \hat{\Phi}, \hat{\Psi})$, the frozen rank-copula maps $\hat{F}_j$ from (A19), and the frozen $V0$ covariate residualization are held constant, and a follow-up record is projected, never re-discovered. Concretely, a V_t score is obtained by (i) orienting and copula-transforming each gaussianized cell onto the $V0$ z-scale via $\hat{F}_j$; (ii) applying the frozen-$V0$ covariate residualization (age-spline, sex, education, site) with the visit's covariates; and (iii) projecting onto the fixed $\hat{\Lambda}, \hat{\Phi}, \hat{\sigma}$ using the EAP map (A28) for the continuous block and the corresponding full-N projection for the explicit block. As a validity check, scoring $V0$ through this path reproduces the baseline coordinates at Pearson $r = 0.99$–1.00 across the continuous axes, so $V0$, $V1$ and $V2$ live on a common scale (panel of 16,241 visit-records: $V0$ 9,013, $V1$ 4,270, $V2$ 2,958).

C.2 Longitudinal measurement invariance

Re-fitting the simple-structure backbone per visit (scale-invariant by design) and computing the Tucker congruence of the primary loadings against $V0$, every backbone axis was invariant at both follow-ups (minimum congruence φ: sleep 0.996, cognition 0.995, overall severity 0.991, metabolic 0.988, inflammatory 0.974; Extended Data Fig. F4). Inflammatory load, only partially invariant on a plain Gaussian map, was invariant here. Coordinate change can therefore be read as patient change rather than instrument drift.

C.3 Trait/state decomposition

For each axis we fit a measurement-error random-intercept model across visits. Writing x_{it} for patient i's coordinate at visit t, with known baseline measurement variance s_i^2 (the relevant diagonal entry of S_i),

$$x_{it} = \underbrace{\gamma_t}_{\text{visit fixed effect}} + \underbrace{u_i}_{\text{trait}} + \underbrace{e_{it}}_{\text{state}} + \underbrace{m_{it}}_{\text{measurement}}, \quad u_i \sim \mathcal{N}(0, \tau^2),\ e_{it} \sim \mathcal{N}(0, \omega^2),\ m_{it} \sim \mathcal{N}(0, s_i^2), \tag{C32}$$

so that plugging the known s_i^2 prevents a low-reliability axis from masquerading as state, and the visit fixed effects γ_t prevent a shared population trajectory from being miscounted as individual instability. The trait fraction is the intraclass correlation

$$\text{ICC} = \frac{\tau^2}{\tau^2 + \omega^2}, \tag{C33}$$

estimated as a posterior with 94% credible interval. The biological and cognitive axes were trait (metabolic 0.91, substance 0.88, cognition 0.70, inflammatory 0.62); the symptom axes were state (sleep 0.47, suicidality 0.42, developmental risk 0.39); general burden was trait by rank (0.62). The population shift $\gamma_{V2} - \gamma_{V0}$ was strongly negative on severity (-0.46) and suicidality (-0.84) and near zero on biology, the formal statement of "the cohort improves on symptoms while individual biological rank is held." A complementary geometric route confirmed that the archetype weight vector moved little (median cosine similarity 0.90 over $V0 \to V2$ completers); the two routes agree (Spearman of per-axis reliable-change rate against ICC $\rho = -0.27$, the correct sign). Developmental risk scores as state partly because childhood-adversity reports carry recall noise rather than true change; mania is data-limited (two indicators) and its trait/state verdict is not load-bearing.

D Prognosis: errors-in-variables incremental validity

D.1 Errors-in-variables Bayesian GLM and the nested ladder

The map coordinates are estimated, not observed, so a naive regression on $\hat{f}_i$ would suffer attenuation. We use an errors-in-variables (EIV) Bayesian generalized linear model that treats the latent coordinate as a parameter with a measurement model attached. For outcome y_i (functioning EGF or severity CGI-S at two years, on its native scale) and design W_i,

$$\theta_{id} \sim \mathcal{N}(\hat{f}_{id}, s_{id}^2), \qquad g(\mathbb{E}[y_i]) = \beta_0 + \beta_W^\top W_i + \beta_\theta^\top \theta_i, \tag{D34}$$

where s_{id} is the per-patient posterior standard deviation from (A28), so the regression "knows" how precisely each coordinate is measured. We compare a fixed ladder of designs W: a reference $\mathrm{R}_{3y} = \{\text{age, sex, baseline outcome, baseline severity}\}$; then +DSM-5 subtype, +map (continuous coordinates, the eight specific axes, the four archetypes, or a tessellation), or +both. The estimand is incremental: does the map add *beyond* diagnosis, current severity and the baseline value of the outcome.

D.2 Incremental value, operative K, and discrimination

Incremental value is the held-out expected log predictive density, estimated by Pareto-smoothed importance-sampling leave-one-out [44]:

$$\Delta\mathrm{ELPD} = \widehat{\mathrm{elpd}}_{\mathrm{loo}}(\text{model}) - \widehat{\mathrm{elpd}}_{\mathrm{loo}}(\mathrm{R}_{3y}), \tag{D35}$$

declared *predictive* when it exceeds twice its standard error. For functioning (N = 2,114): archetypes +59, eight specific axes +37, every tessellation $\approx +20$, the isolated durable-biology axes +2 (ambiguous). The continuous and archetype encodings strictly dominate any hard partition, so the operative-K selector returns *none*—the actionable object is a continuous position. For severity the increments are small and ambiguous (autoregression-saturated). Head-to-head, map and DSM-5 are co-informative (functioning: +map 22, +DSM-5 29, +both 67). For binary endpoints we also report the change in AUC under five-fold cross-validation; functional-remission AUC rose only $0.745 \to 0.756$ (+0.011), the signature of a strong group-level signal that does not translate into a large individual gain. The two-year functional-remission gradient across archetypes (27% biological corner to 60% low-burden) was a within-diagnosis effect: the rank held inside each cohort (biological-vs-low-burden odds ratio ≈ 4), direct standardization to a common cohort mix moved it ≈ 6%, and the corner-by-cohort interaction was non-significant.

D.3 Robustness

Stressing the predictive winner (the archetypes) on functioning: the signal survived inverse-probability-of-attrition weighting (+53), dropping DR or SZ (+56, +59), and vanished under a permutation null (-2, correctly); it weakened when bipolar disorder was removed (+6), so the increment is carried by the episodic, open-course cohorts. The isolated durable-axis block, which cleared the bar on an earlier plain-Gaussian map, did *not* survive here; the robust predictive object is the fuller archetype phenotype (the biological corner), reported as an honest, map-specific shift.

D.4 Representation sufficiency benchmark

Because the nine-dimension map is a $\approx$ 15-fold compression of the 143 raw indicators, the honest question is sufficiency, not supremacy. Under one fixed regularized gradient-boosting classifier with identical cross-validation folds, we compared three feature sets—reference (DSM-5 + severity + baseline GAF), reference + map (coordinates, uncertainty and archetypes), and reference + the 143 raw indicators—predicting recovery and deterioration. The map was *sufficient* for deterioration (AUC tie) and *near-sufficient* for recovery (raw +0.04 AUC); a TreeSHAP attribution showed that 97% of the raw recovery signal already lives *within* the nine modelled factors (top drivers: CRP, BMI/HbA1c/lipids, verbal memory and processing speed, nicotine dependence, childhood adversity), so the residual is within-factor compression, not a missing axis (Extended Data Fig. F5). The map also transported across held-out cohorts at least as well as the raw indicators for deterioration.

E Treatment: an identification-first causal pipeline

This baseline cohort has no randomization (the trial-arm field is a DSM-5 subtype), so treatment *selection*—who should get which drug—is a counterfactual question observational treatment-as-usual cannot answer. The pipeline therefore *bounds* what the map licenses and *defends* the prognostic finding, rather than claiming a treatment effect.

E.1 Exposures, overlap and weighting

Treatment exposures, recovered from the per-cohort source dictionaries, were harmonized to common drug-class indicators. For each treatment contrast we estimate a propensity $e(v_i) = P(\text{treat} \mid v_i)$ on a design v_i that includes severity, diagnosis, demographics *and* the nine coordinates [47], gate on common support (retain the region where both arms have mass), and reweight by stabilized inverse-probability-of-treatment weights [48]

$$w_i = \frac{P(\text{treat})}{e(v_i)} \text{ (treated)}, \qquad w_i = \frac{1 - P(\text{treat})}{1 - e(v_i)} \text{ (control)}, \tag{E36}$$

checking covariate balance by the post-weighting maximum standardized mean difference. Lithium-in-BP balanced to a maximum SMD of 0.008 (near-complete overlap 0.997); antipsychotic-in-BP balanced to 0.079; clozapine-in-SZ retained residual imbalance (SMD 0.335), the signature of channelling.

E.2 Doubly-robust moderation, the E-value and the minimum detectable effect

The estimand of interest is *moderation*: does a durable axis or archetype change *who* benefits. We fit a doubly-robust errors-in-variables outcome model with a treatment-by-coordinate interaction (the EIV term as in (D34)), reporting the per-axis interaction credible interval and the held-out ΔELPD of adding the interaction. Each average treatment effect carries an E-value for unmeasured confounding [49]—the minimum strength of association an unmeasured confounder would need, with both treatment and outcome, to explain away the estimate—and, crucially, a *minimum detectable effect* (MDE): the smallest interaction the design resolves at 80% power. The MDE is what makes a null *bounded* rather than blind.

- **Lithium-BP** (cleanest): average effect on functioning -0.003 (E-value 1.06), with per-axis interactions well inside an MDE of 0.19–0.22 SD—a *well-identified, bounded null*. The design could have resolved a meaningful interaction and did not.
- **Antipsychotic-BP**: a confounded *average* effect (-0.231, E-value 1.77) but *no reliable moderation*—the interaction ΔELPD band spans zero and the two map encodings disagree on which axis carries it (the durable representation points at metabolic, the archetype representation at the biological corner), textbook false-positive behaviour.
- **Clozapine-SZ**: channelled and underpowered (MDE 0.37–0.67 SD), a non-decisive null rather than a clean one.

Average treatment effects were uniformly confounding-fragile (E-values 1.1–1.8).

E.3 The map is not a treatment proxy (defending the prognosis)

The most damaging alternative to the prognostic finding is that the biological corner simply received different drugs that drove the outcome. We refit the functioning prognosis with and without the harmonized drug-class exposures, both unweighted and under attrition weighting. The archetype carrier of the forecast survived (β $0.164 \to 0.156$ unweighted, attenuation 4.7%; $0.147 \to 0.141$ under attrition weighting, 3.9%; both credible intervals excluding zero), so the functional forecast is not merely unmodelled treatment. (The isolated durable trio did not survive unweighted, consistent with the prognosis annex; its metabolic axis sharpened to exclude zero only under weighting.) The adjustment is for baseline/lifetime (BP) or current (SZ/DR) drug-class exposure, not a marginal structural model over time-varying treatment.

E.4 Treatment-course atlas

Even without selecting a drug, the baseline archetype *describes* two-year treatment course. Per corner (pooled BP+SZ, Wilson intervals): treatment-resistance 43% (biological) to 20% (low-burden); CGI response 46% to 59%; significant side-effects 23% to 11%. The corner added beyond baseline severity, substance comorbidity and demographics (likelihood-ratio $p \leq 3 \times 10^{-3}$ for all three endpoints), the gradient was within-cohort (composition share $\leq 6\%$, corner-by-cohort interaction non-significant), and held-out ΔELPD was $+20/+16/+10$. Individual discrimination was modest: under a permutation null on the held-out ΔAUC, response ($p = 0.015$) and side-effects ($p = 0.005$) cleared, while treatment-resistance—the steepest gradient—was AUC-marginal ($p = 0.185$), the same group-versus-individual collapse seen for prognosis. The atlas is therefore a monitoring signal (which phenotype to watch more closely), never a prescription.

F Extended Data figures and the claims ledger

F.1 Extended Data figures

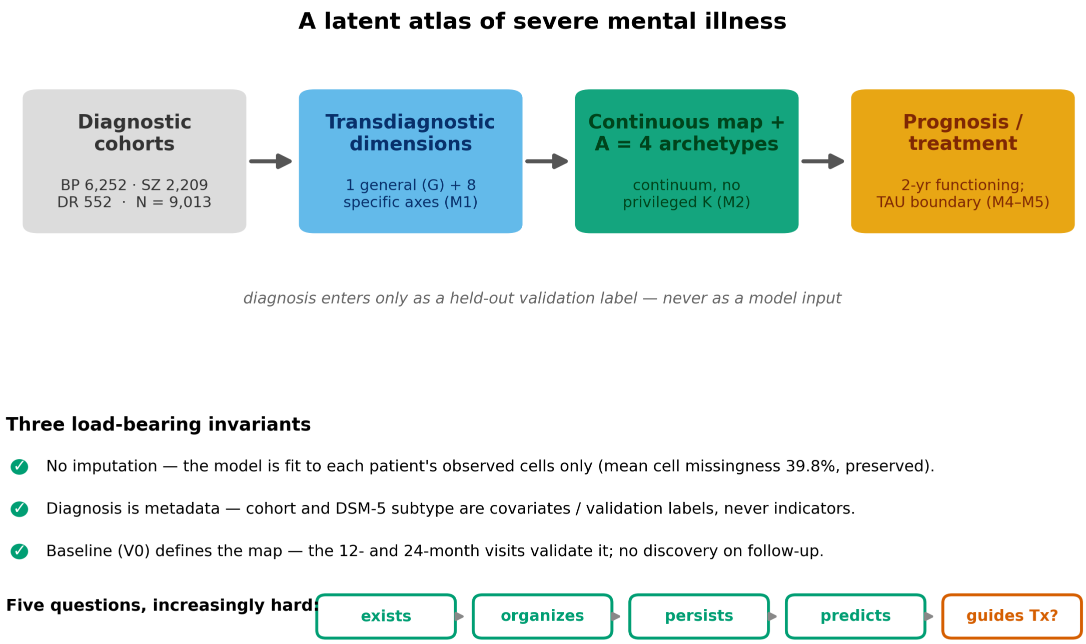

Figure F1: **Study design, invariants and the five questions.** The harmonized three-cohort FACE baseline (bipolar disorder, schizophrenia, depression; $N = 9{,}013$) is reorganized around diagnosis-agnostic axes by one global, missingness-aware Bayesian bifactor / exploratory structural-equation model fit to each patient's observed cells only, under three load-bearing invariants (no imputation; diagnosis as metadata; baseline defines the map and follow-up validates it). The fitted map is then interrogated through five dependent, increasingly demanding questions—exists, organizes, persists, predicts, guides treatment—with positive and null results reported alike.

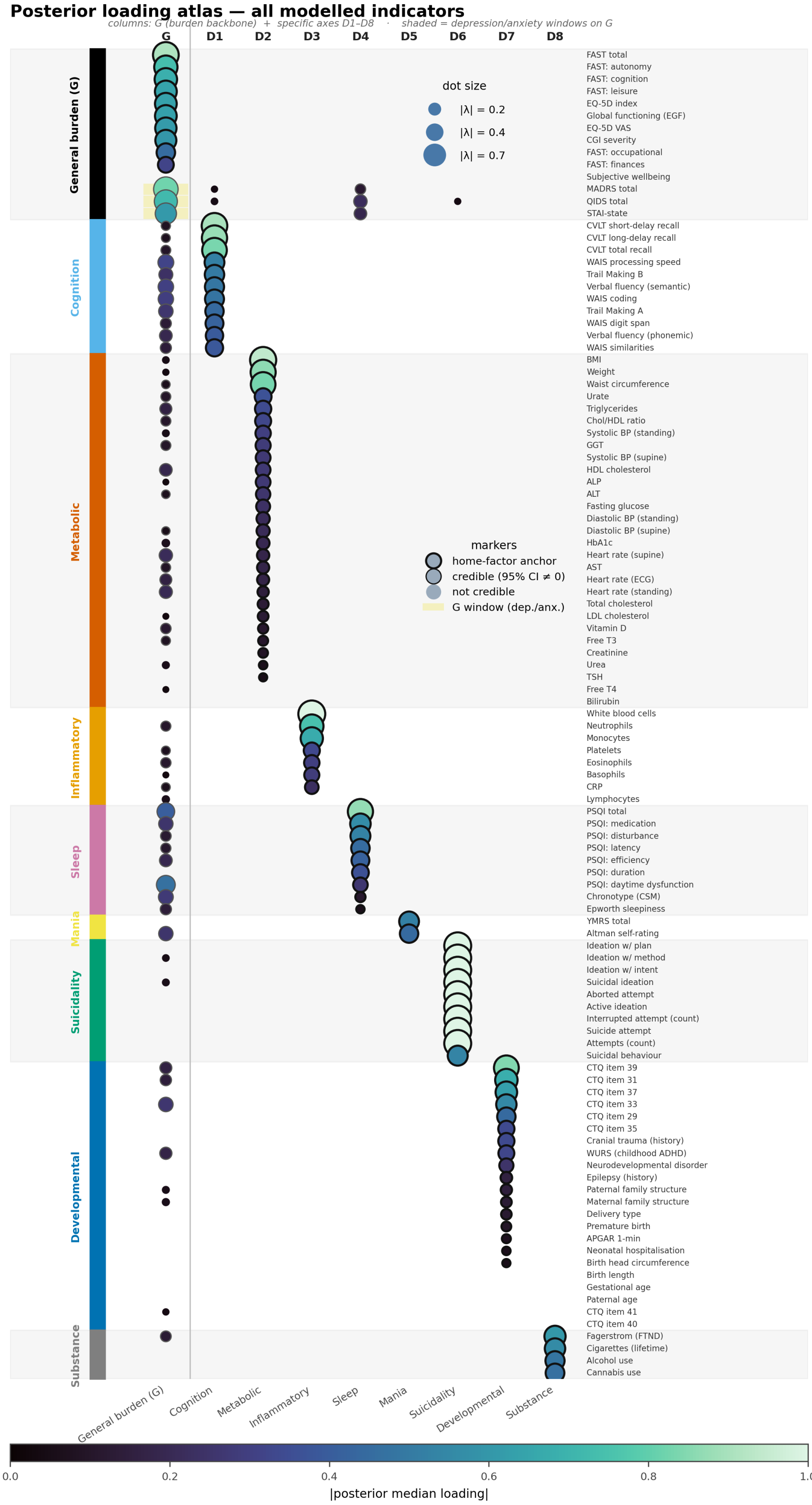

Figure F2: **The complete posterior loading atlas.** As Figure 2a but showing *every* modelled indicator (all 109 continuous and explicit items) grouped by home factor, with no per-block cap. Dot size and colour encode the absolute posterior median loading; outlined dots have a 95% credible interval excluding zero (heavier ring = home-factor anchor). The diagonal block structure—each clinical block loading cleanly on its own specific axis, with a sparse general-burden (G) column—is the full anatomy of the measurement model summarised in the main-text Figure 2.

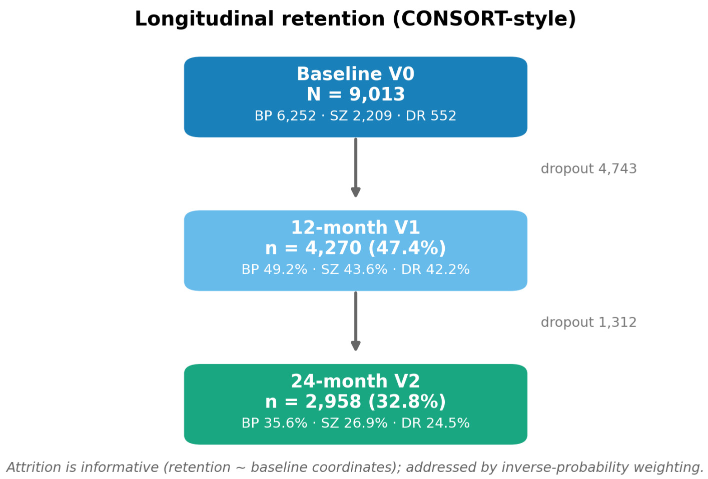

Figure F3: **Longitudinal retention.** Patients assessed at baseline and retained at the 12- and 24-month follow-up visits, overall and by cohort. Attrition is informative (retention depends on baseline coordinates) and is addressed throughout by inverse-probability weighting.

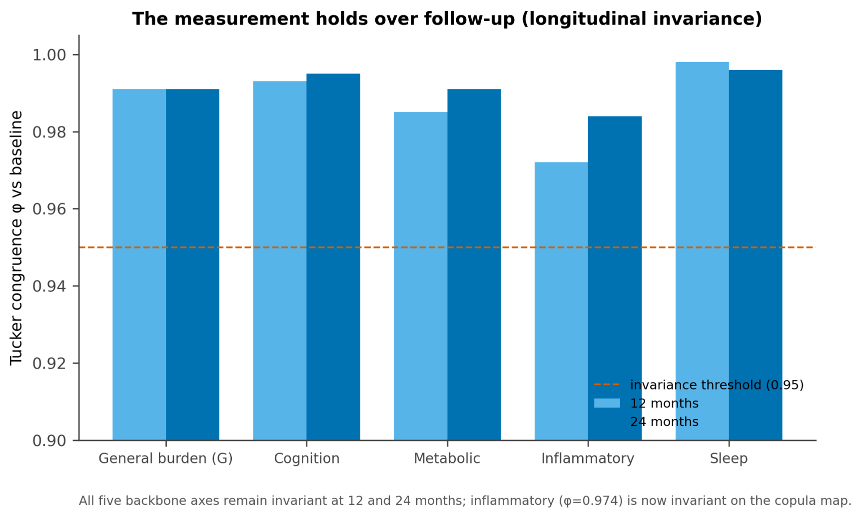

Figure F4: **Longitudinal measurement invariance.** Tucker congruence of each backbone axis against baseline at 12 and 24 months. All five axes remain invariant ($\varphi \geq 0.95$); inflammatory load (only partially invariant on a plain Gaussian map) is invariant here.

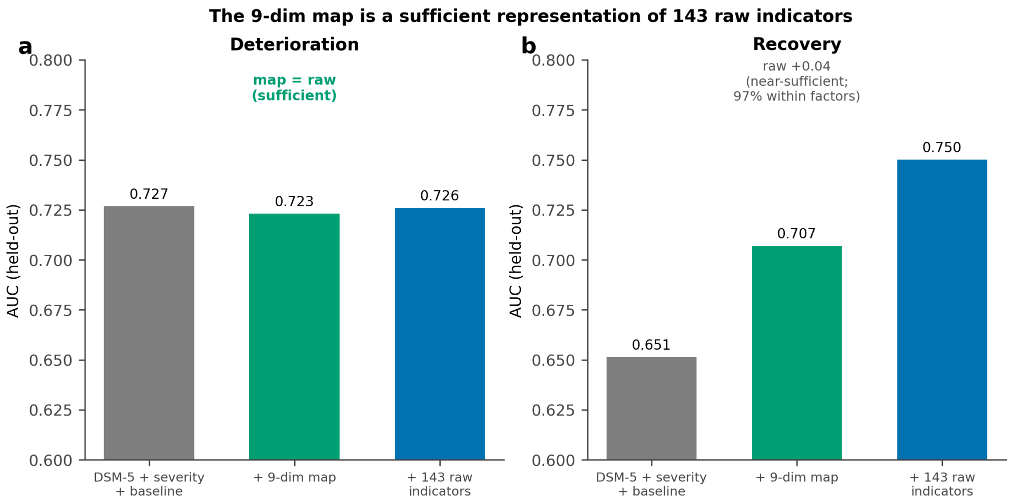

Figure F5: **The nine-dimension map is a sufficient representation of the** 143 **raw indicators.** Held-out AUC for a matched gradient-boosting classifier predicting two-year deterioration and recovery, using the reference (DSM-5 + severity + baseline), the reference + the nine-dimension map, or the reference + all 143 raw indicators. The map equals the raw indicators for deterioration (sufficient) and trails by ≈ 0.04 AUC for recovery (near-sufficient); 97% of the raw recovery signal lies within the nine factors.

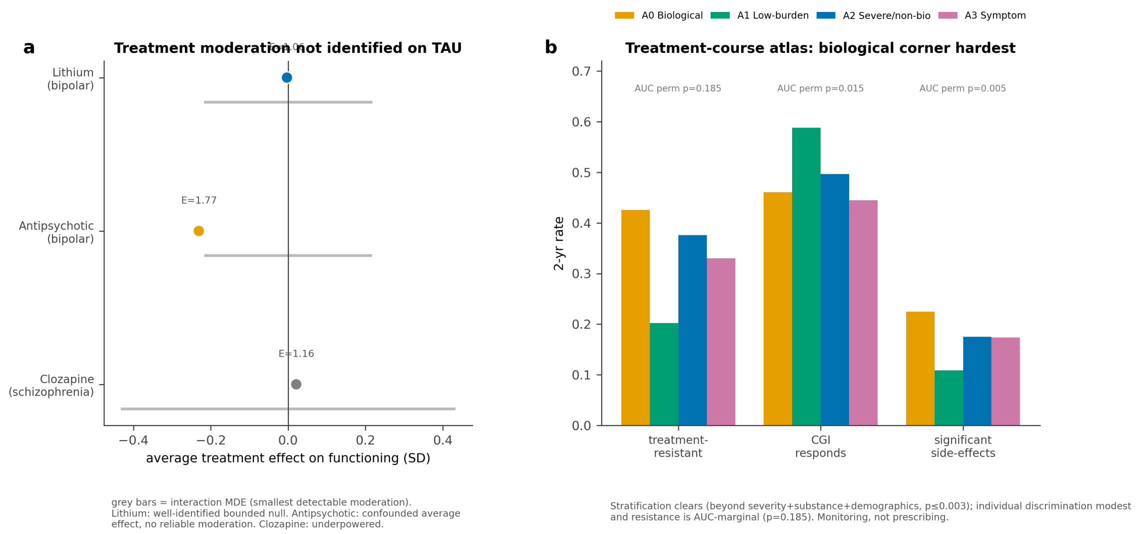

Figure F6: **Treatment moderation is not identified on observational treatment-as-usual.** (**a**) Average treatment effect on functioning with E-value and the interaction minimum-detectable-effect band (grey). Lithium in bipolar disorder is a well-identified, bounded null; antipsychotic is a confounded average effect with no reliable moderation; clozapine is channelled and underpowered. (**b**) Treatment-course atlas: the biological corner carries roughly twice the resistance and side-effect burden of the low-burden corner; stratification clears beyond confounders but individual discrimination is modest (resistance AUC-marginal under permutation).

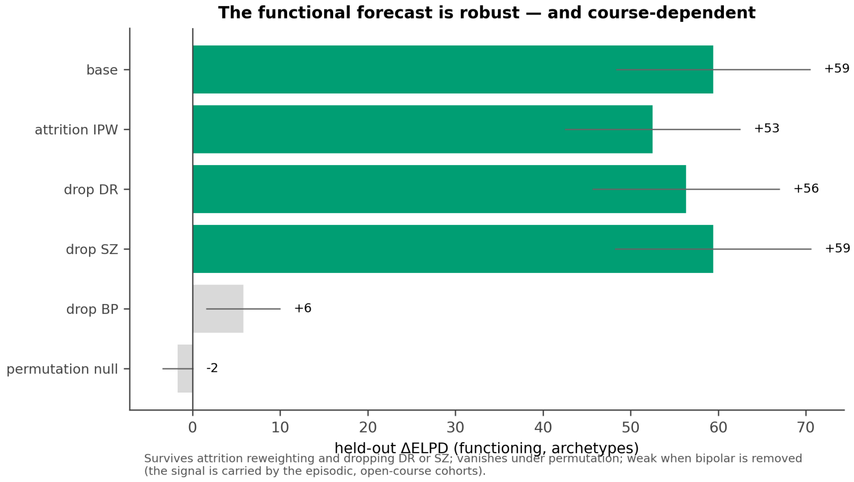

Figure F7: **The functional forecast is robust and course-dependent.** Held-out incremental value of the archetype phenotype for two-year functioning under attrition reweighting, cohort hold-outs and a permutation null. The signal survives attrition and dropping either DR or SZ, vanishes under permutation, and weakens when bipolar disorder is removed—it is carried by the episodic, open-course cohorts.

F.2 The claims ledger

Table F3 states, side by side, what the study claims and what it deliberately does not—the spine of the Limitations and of any response to reviewers.

Table F3: **What we claim, and what we do not.**

We claim	**We do not claim**
A certified nine-dimension transdiagnostic measurement map (internal validity).	Natural biotypes or subtypes; that no biotype exists in any measurement space.
Metabolic/inflammatory load *largely independent* of a general functional-burden axis (quantified, with boundary conditions).	Strict statistical orthogonality, or independence from *being ill* (case–control elevations).
A graded continuum that is, descriptively, tighter than DSM-5.	That the regions are clinically superior to DSM-5 beyond the modest prognostic increment shown.
Temporal coherence: durable biology, moving symptoms.	That developmental risk is genuinely "state" (it partly reflects recall noise).
A modest, *group-level* incremental prognosis for *functioning*, carried by the biological phenotype, course-dependent.	A large individual-risk gain; prediction of *recorded* relapse/hospitalization events; a deployable individual rule.
A well-identified null for lithium moderation; a testable antipsychotic hypothesis.	That the map guides treatment selection, or any causal treatment effect on treatment-as-usual.
Convergent validity with the immuno-metabolic literature.	External or out-of-sample validation (none yet).

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